

A rapid return to normal: temporal gene expression patterns following cold exposure in the bumble bee *Bombus impatiens*

Kelton M. Verble^{1,*}, Ellen C. Keaveny², Sarthok R. Rahman¹, Matthew J. Jenny¹, Michael E. Dillon², Jeffrey D. Lozier¹

¹Department of Biological Sciences, The University of Alabama, Tuscaloosa, AL 35487 USA

²Department of Zoology & Physiology and Program in Ecology and Evolution, University of Wyoming, Laramie, WY 82072 USA

*Corresponding Author: Kelton M. Verble

Department of Biological Sciences, The University of Alabama

Box 870344, Tuscaloosa, AL 35487

(205) 348-1807

kmverble@crimson.ua.edu

Keywords: pollinator, climate, chill coma, metabolism, TagSeq, differential gene expression

SUMMARY STATEMENT

Following acute cold exposure, bumble bees mount a rapid, yet short-lived, transcriptional response characterized by changes in gene expression patterns that influence metabolic shifts.

ABSTRACT

Bumble bees are common in cooler climates and many species likely experience periodic exposure to very cold temperatures, but little is known about the temporal dynamics of cold response mechanisms following chill exposure, especially how persistent effects of cold exposure may facilitate tolerance of future events. To investigate molecular processes involved in the temporal response by bumble bees to acute cold exposure, we compared mRNA transcript abundances in *Bombus impatiens* workers exposed to 0°C for 75 minutes (inducing chill coma) and control bees maintained at a constant ambient temperature (28°C). We sequenced the 3' end of mRNA transcripts (TagSeq) to quantify gene expression in thoracic tissue of bees at several time points (0 minutes, 10 minutes, 30 minutes, 120 minutes, and 720 minutes) following cold exposure. Significant differences from control bees were only detectable

within 30 minutes after the treatment, with most occurring at the 10-minute recovery time point. Genes associated with gluconeogenesis and glycolysis were most notably upregulated, while genes related to lipid and purine metabolism were downregulated. The observed patterns of expression indicate a rapid recovery after chill coma, suggesting an acute differential transcriptional response during recovery from chill coma and return to baseline expression levels within an hour with no long-term gene expression markers of this cold exposure. Our work highlights the functions and pathways important for acute cold recovery, provides an estimated time frame for recovery from cold exposure in bumble bees, and suggests that cold-hardening may be less important for these heterothermic insects.

INTRODUCTION

The ability to mount an appropriate physiological response to challenging abiotic conditions can determine the functional capacity and viability of organisms that encounter variable environments (West-Eberhard, 1989; Whitman & Agrawal, 2009). Under dynamic environmental conditions, individuals can benefit from flexible phenotypic responses that can match fluctuations in external stimuli (Lande, 2009; Schlichting & Pigliucci, 1993; Stomp et al., 2008). At the molecular level, individuals can exhibit such flexibility through differential gene expression, where transcription of genes can be modified in response to environmental stressors, such as extreme temperature changes (Baniwal et al., 2004; DeSalvo et al., 2008; Schoville et al., 2012; Shearer et al., 2016; Sonna et al., 2002). Given ongoing temperature shifts associated with global climate change, there is substantial interest in characterizing the physiological and molecular responses of extreme thermal events (Buckley & Somero, 2009; DeSalvo et al., 2008; Harvey et al., 2023; Heikkila, 2017; Sonna et al., 2002).

For insects, thermal tolerance is a driving factor that shapes species distributions and life history strategies (Bale, 2002; Kellermann et al., 2012; Oyen et al., 2016; Stange & Ayres, 2010; Sunday et al., 2011), and gene expression modification is a key component of responses to temperature fluctuations (Des Marteaux et al., 2017; Kim et al., 2006; Lubawy et al., 2022; Malmendal et al., 2006; Sinclair et al., 2007; Teets & Denlinger, 2013; Xu et al., 2017; Zhao & Jones, 2012). Climate change has driven substantial research on tolerance of warming conditions, however, understanding the mechanisms underlying cold tolerance may be as important as studying upper thermal limits, as exposure to cold events (Teets et al. 2023) and the frequency and severity of extreme weather events (Ummenhofer & Meehl, 2017) will continue despite rising mean temperatures. Interestingly, while the upper range for heat tolerance (i.e., the maximum critical thermal limit, or CT_{max}) is relatively similar across insects

(~40°C; Neven, 2000) and generally stable within species (Sunday et al., 2012), species and populations can vary in cold tolerance (Bowler & Terblanche, 2008; Oyen et al., 2016; Pimsler et al., 2020; Sunday et al., 2014). For example, many insects exhibit large-scale gene expression alterations that occur during acclimation to gradual seasonal changes, such as the weeks or months leading to diapause (Costa et al., 2020; Denlinger et al., 2001; MacRae, 2010), although these drastic alterations can also occur in as little as six days (MacMillan et al., 2016a). Sudden thermal stress and cold hardening tend to elicit more subtle and immediate transcriptional responses over minutes to hours (Teets et al., 2020; Teets et al., 2023). Such immediate responses can be crucial for species with life history strategies that expose individuals to more rapid shifts in temperature (Bale et al., 2002; Kingsolver et al., 2011; Sinclair, 2001), such as species active at different times of day or during seasonal transitions (Colinet et al., 2015). Several mechanisms that can influence cold tolerance are similar to those observed in response to heat exposure, such as the increase in metabolism (particularly glycolysis) and alterations in the expression of heat shock proteins (Teets & Denlinger, 2013). The production of cryoprotectants, shifts in cellular membrane composition and in ion transport, and cytoskeletal restructuring are some of the other mechanisms that may be controlled by rapid molecular changes to protect against rapid temperature reductions (MacMillan et al., 2014; Overgaard & MacMillan, 2017; Robertson et al., 2017; Teets & Denlinger, 2013). Such rapid responses of enhanced cold tolerance are protective but energetically expensive (Chen and Walker, 1994), and benefits can rapidly dissipate (Wang & Kang, 2003). Some work has evaluated shifts in gene expression for insects subjected to acute cold (Sinclair et al., 2007; Teets et al., 2012; Zhang et al., 2011), but for most species we still require a better characterization of transcriptomic profiles immediately after extreme thermal events to better understand both how organisms maintain or restore physiological functions and whether effects of thermal stress persist beyond the period of exposure.

Bumble bees (Hymenoptera: Apidae: *Bombus* Latreille 1802) provide an interesting model organism with which to evaluate responses to cold stress. *Bombus* evolution and biogeography has been shaped by global cooling periods (Hines, 2008; Martinet et al., 2018; Williams et al., 2018), and species diversity is high in temperate and cold (e.g., montane, arctic) environments. Although bumble bees establish thermally controlled colonies in cavities during the growing season, individuals of different castes can face varying levels of cold exposure risk while outside the nest. For example, queens emerging in the spring may spend weeks unprotected above ground, risking exposure to early spring snowstorms before settling underground in their established colonies (Makinson et al., 2019). Workers can forage >1km from the colony and occasionally remain outside of the colony overnight, increasing

their risk of experiencing extreme cold (Free, 1955; Müller & Schmid-Hempel, 1993; Walther-Hellwig & Frankl, 2000). To promote heat regulation and retention in the face of thermal challenges, bumble bees are facultatively endothermic insects (Heinrich, 2004), with morphological features such as large body size and dense pile that aid in temperature regulation (Heinrich, 2004; Maebe, Hart, et al., 2021; Peters et al., 2016). Bumble bees can thus maintain elevated body temperatures in conditions that are often too cold for many ectothermic insects to function and are capable of activities earlier in the day or season compared to other pollinators (Goulson, 2010). The metabolic powerhouse of heat production in bumble bees lies in their thoracic muscles (Heinrich, 2004), which not only power flight but also allow bumble bees to warm up prior to flight and for other activities (Esch et al., 1991; Heinrich, 1993; Heinrich & Esch, 1994). However, when cooled sufficiently, bees and other insects reach their critical thermal minimum (CT_{min}), when loss of central nervous system function and impairment of peripheral neuromuscular junctions leads to a state of arrested movement known as chill coma (Andersen et al., 2015; MacMillan & Sinclair, 2011). Previous studies have demonstrated regional differences in bumble bee CT_{min} both among species (Maebe, De Baets, et al., 2021; Oyen et al., 2016) and populations (Pimsler et al., 2020). For example, *B. vosnesenskii* workers reared from queens sampled across an elevational and latitudinal gradient had CT_{min} strongly correlated with temperatures at queen collection sites, whereas critical thermal maxima (CT_{max}) values were relatively invariable (Pimsler et al., 2020). Bees from different regions displayed variations in gene expression in response to cold stress, while responses to heat stress were strong but conserved among regions (Pimsler et al., 2020). Because bumble bees are intolerant of freezing (Keaveny et al., 2022), and must navigate variable conditions without approaching their relatively high freezing point to prevent cold injury and death, molecular mechanisms for tolerating periods of cold stress may be an important component of bumble bee environmental adaptation.

In this study, we further investigate differential gene expression responses associated with cold stress in the common eastern bumble bee, *Bombus impatiens* Cresson 1863, a wide-ranging generalist pollinator found across eastern and central North America (Williams et al., 2014). We aimed to test if acute cold exposure has lasting effects on gene expression or whether the high metabolic rate (and, therefore, fast metabolic turnover) of facultatively endothermic bumble bees leads to a rapid and short-lived molecular response. We induced chill coma and assessed differences in partial mRNA transcripts using 3' tag RNA sequencing (TagSeq; Lohman et al., 2016; Meyer et al., 2011) at pre-determined time points between treatment and control bees to capture potential effects immediately after cold exposure and up to several hours after recovery from chill coma. If differentially expressed genes are detectable

earlier in a sampling time series, we hypothesize that such genes indicate immediate and short-term *B. impatiens* chill coma recovery processes. Alternatively, if differentially expressed genes are important for long term recovery and potentially beneficial for protection from subsequent cold exposures, expression of such markers should persist and be detectable in later sampling periods.

MATERIALS AND METHODS

Animals

Three commercial *B. impatiens* colonies (Koppert Biological Systems, Howell, MI, USA) were maintained at 28°C with 60% RH on a 12:12 light:dark cycle in an incubator (Model #I36VL, Percival Scientific, Perry, IA, USA) in Laramie, WY, USA. They were fed nectar ad libitum via a wick and reservoir provided by the supplier in addition to 1 tbsp. ground fresh frozen pollen (Prairie River Honey Farm, Grand Island, NE, USA) every other day. Seventy-two hours prior to experiments, twenty workers from each of the commercial colonies were randomly selected and placed in colony-specific control and treatment chambers (Figure 1A) and were fed a pollen ball (mix of ground pollen and nectar) and nectar (1:1:2 sucrose, invert sugar, and distilled water, with <0.1% of sorbic acid, Honey B Healthy [Honey B Healthy, Cumberland, MD], Amino-B Booster [Honey B Healthy, Cumberland, MD], and citric acid added) ad libitum through a wick inserted into a nectar-filled vial attached to the chamber wall. A small piece of comb from the respective natal commercial colony was added to each microcolony. The six microcolonies were returned to the incubator after changing the light:dark cycle to 16:8 L:D where they remained for a 72-hour settlement period.

Experimental Protocol and Sampling

After the 72-hour settlement period, we placed three microcolonies (one from each commercial colony) into a small plastic container (to ensure that the microcolonies stayed dry) that was submerged into an ice bath inside of a cooler (ambient air measured 0°C) for 75 minutes (Figure 1B). Conditions for the remaining three microcolonies were held constant in the 28 °C incubator to serve as the control. At the end of the 75-minute cold exposure, three bees from each of the six microcolonies (three control and three treatment microcolonies) were randomly selected and flash-frozen (time 0 after cold exposure, n=18, referred to as 00-min), before returning the cold-treated microcolonies with all bees still ventrally curled from chill-coma to the incubator. Once in the incubator, we started a timer in the view of a video camera (Go-Pro, San Mateo, CA, USA) mounted above the cold-treated microcolonies which allowed for

measurement of recovery time by monitoring movements of individual bees. All bees sampled at 00-min were in chill coma whereas all other bees sampled were fully recovered from chill coma. Three bees from each of the six microcolonies were flash-frozen at 10 minutes, 30 minutes, 2 hours, and 12 hours, as above (referred to as 10-min, 30-min, 120-min, and 720-min sampling periods) after the cold exposure ended (n= 9 bees per sampling time, per treatment type; Figure 1). All bees were stored at -80°C until preparation for RNA extraction. To split bees, a metal stage was positioned on top of a block of dry ice on a sterilized surface. One bee was removed from the cryovial, weighed, placed on the stage, and cut along a sagittal plane using a sterile razor and forceps. There were no significant differences in mass among treatments or timepoints (ANOVA, all p-values > 0.148). Each half was placed into separate cryovials and returned to -80°C, with one half for RNA isolation and the other reserved for archiving and possible follow-up experiments.

RNA Sequencing and Gene Expression Analysis

Due to the significance of flight muscle in thermoregulation (see Introduction), RNA was isolated from thoracic tissue (Invitrogen RNAqueous- Micro Kit, Thermo Fisher Scientific) following the manufacturer's instructions, with the extraction order of samples randomly selected across colonies and treatments to minimize potential batch effects. Aliquots of RNA at 10 ng/μL were sent on dry ice to the University of Texas at Austin Genomic Sequencing and Analysis Facility (Austin, TX, USA) for quality control and 3' Tag-based RNA sequencing (TagSeq) library preparation and sequencing (Lohman et al., 2016; Meyer et al., 2011). Details are provided in Lohman et al. 2016 but, briefly, TagSeq is a reduced representation approach for RNA sequencing of the 3' ends of transcripts. cDNA libraries are prepared by isolating mRNA and subjecting the transcripts to a heat-based fragmentation to reduce length bias. A unique oligo-dT primer targeting the 3' end of fragmented transcripts is used for first strand synthesis, and the libraries are PCR amplified for sequencing (Lohman et al., 2016; Meyer et al., 2011). Sequencing (100bp, single-end) was conducted across two lanes on a NovaSeq 6000 SR100 (Illumina) flowcell. Samples used in analyses were sequenced to a mean of 12.85 million ($\pm$ 2.3 million SD) reads per sample; a small number of samples that did not produce successful libraries or meet our sequence data threshold (< 8 million reads) were excluded (see Table.SI.1 for final sample sizes per colony per treatment). The final number of samples used in the analyses per timepoint, per treatment are as follows: 00-min control bees = 4, 00-min treatment bees = 8; 10-min control bees = 8, 10-min treatment bees = 7; 30-min control bees = 8, 30-min treatment bees = 8; 120-min control bees = 8, 120-min treatment bees = 9; 720-min control bees = 7, 720-min control bees = 7).

Quality assessment of reads was conducted using FastQC v0.11.9 (Andrews, 2010). The two lanes were concatenated into a single FASTQ file per sample. As recommended for TagSeq data, the *tagseq_clipper.pl* script from the “tag-based_RNAseq” pipeline by Mikhail Matz (https://github.com/z0on/tag-based_RNAseq, pipeline version December 29, 2017) with cutadapt v1.18 (Martin, 2011) was used to clean and deduplicate reads with the following command for each sample: `tagseq_clipper.pl {SAMPLE_ID}.fq | cutadapt -a AAAAAAAAA -a AGATCGG -q 15 -m 25 -o {SAMPLE_ID}.trim`. We used STAR v2.7.6a (Dobin et al., 2013) to align the cleaned reads to the *B. impatiens* genome (BIMP2.0, GCA_000188095.2) (Sadd et al., 2015) and generate transcript counts per locus (sense strand). The R v4.0.3 (R Core Team 2020) package DESeq2 v1.30.1 (Love et al., 2014) was used for differential expression analysis from the transcript counts matrix. The DESeq() function from DESeq2 estimates size factors while accounting for differences in sequencing depth, calculates dispersion estimates for how much the counts of each sample will vary around expected values, and fits dispersion estimates to a generalized linear model, which provides the estimates of the log fold changes between contrasts. Because variance was substantially different among time points, we elected to analyze each time point separately for the DESeq2 analyses. To pinpoint the time at which gene expression differences between treatment and control bees become evident and/or dissipate, we analyzed models to test for a treatment effect at each time point, with source colony specified as a covariate. DESeq2 uses Wald testing to determine p-values in the gene expression analysis, and the Benjamini-Hochberg method (Benjamini & Hochberg, 1995) to account for multiple testing and provide an adjusted p-value. Our significance threshold for significant differentially expressed genes (DEGs) was an adjusted p-value ≤ 0.05 . Plots and figures were created using DESeq2, the R package VennDiagram v1.7.3 (Chen & Boutros, 2011), and ggplot2 v3.3.6 (Wickham, 2011). The R package GeneOverlap v1.34.0 (Shen, 2014) was used to determine if DEG overlaps between time series were greater than expected using Fisher’s exact tests.

Identifying Gene Co-Expression Networks

As a secondary approach to identify genes that respond to cold treatment across time points, we identified co-expression patterns among genes using a weighted gene co-expression network analysis with the R package WGCNA v1.70-3 (Langfelder & Horvath, 2008). We used Pearson correlation coefficients and a scale free network to create a dissimilarity matrix across the entire gene dataset that identified “modules” of co-expressed genes with relative expression patterns that were summarized by module eigengene values. Modules were identified using a dynamic cut tree approach (with minimum

cluster size of 30) on a dendrogram created through hierarchical clustering of the dissimilarity data. The WGCNA analysis is complementary to the DESeq2 analysis above and has the benefit of reducing the number of statistical tests and risk of false positives or negatives associated with differential expression analysis of individual transcripts. To test for associations between module eigengene values and experimental factors we used linear mixed effects models in the R package lme4 (Bates et al., 2014). The full model tested for each module included treatment (cold or control), time point, and a treatment-time interaction, with colony specified as a random effect: [Eigengene ~ Treatment + Time + Treatment*Time + (1|Colony)]. We also tested alternative models without the interaction (Treatment + Time), or with only time (Time Only) or treatment (Treatment Only) as fixed effects. We identified the best fit model for each module using the Akaike Information Criterion (AIC) and only retained modules that had a significant fixed effect of interest determined with the R package lmerTest v3.13 (Kuznetsova et al., 2017) (Satterthwaite approximation, $p\text{-value} \leq 0.05$).

Gene Ontology Enrichment Analysis

We performed gene ontology (GO) enrichment analyses with the R package GOfuncR v1.10.0 (Grote, 2018). Hypergeometric tests were conducted for each time period by comparing a list of DEGs to a background list created using *B. impatiens* specific GO annotations downloaded from the Hymenoptera Genome Database (Walsh et al., 2022). GO terms generated for each gene set of interest were assigned biological process, cellular component, and molecular function ontology categories. Terms that were significant ($p\text{-value} \leq 0.05$) and had a family-wise error rate (FWER) ≤ 0.1 were passed through Revigo (Supek et al., 2011) (web-based interface at <http://revigo.irb.hr/>) to summarize and simplify the GO term list (allowed similarity = small, or 0.5). The R package GOplot v1.0.2 (Walter et al., 2015) was used to visualize the results. For the WGCNA module analysis, the gene sets within each module were evaluated for GO enrichment utilizing the GOfuncR -based approach while maintaining the same criteria and thresholds mentioned in the previous section. Results for the WGCNA modules were further visualized using the R package simplifyEnrichment v1.0.0 (Gu & Hübschmann, 2022), which clusters functional terms in a similarity matrix using the binary cut method. Since gene ontology analyses can provide long lists of functional terms with high levels of redundancy, programs such as simplifyEnrichment attempt to use semantic measures to cluster terms into groups that have high within-group similarity and mutual exclusivity between groups. This can greatly simplify the results of functional analyses and aid biological interpretations. The simplifyGO() function of this package

performs the clustering and produces a heatmap for each defined cluster, along with word cloud annotations comprised of key functional terms associated with each cluster.

RESULTS

Differential Gene Expression and Gene Ontologies

All workers recovered from chill coma within 10 minutes of returning to room temperature (Figure SI.1). Only workers collected at 10-min and 30-min sampling periods after the cold soak showed significant transcriptional responses compared to controls (Figure 2A; Supplemental Material SD.1 and SD.2). In the following, we refer to “upregulated” or “downregulated” transcripts as those with relatively high or low expression, respectively, in cold treatment samples relative to controls. At the 10-min time point, 371 significant DEGs were detected (231 upregulated, 140 downregulated; Figure 2B; SD.1), and 51 DEGs were detected at 30-min (36 upregulated, 15 downregulated) (adjusted p-value < 0.05). The full set of significant genes is presented in Supplementary Material SD.1, but shared genes upregulated in both the 10-min and 30-min time points were *glycogen phosphorylase* (LOC100746613; Figure 2C), *serine/arginine repetitive matrix protein 1* (LOC100744870), *facilitated trehalose transporter Tret1* (LOC100742930), and two *uncharacterized* loci (LOC100741559, LOC105680363), while shared downregulated genes were *phospholipase A1* (LOC100747885; Figure 2C) and *monocarboxylate transporter 13-like* (LOC100748182).

Genes involved in metabolic processes provided the clearest response following recovery from cold exposure. At the 10-min sampling period, eight significant biological process GO terms were identified (Supplemental Material SD.3), primarily associated with carbohydrate synthesis and metabolism (e.g., terms included “hexose metabolic process”, “pyruvate metabolic process”, and “generation of precursor metabolites and energy”; Figure 3). Significantly upregulated DEGs (Table 1; Figure 3) associated with carbohydrate-related GO terms included *phosphoenolpyruvate carboxykinase 1* (LOC105680266), *glycogen phosphorylase* (LOC100746613), *pyruvate kinase* (LOC100743581), *enolase* (LOC100746870), *pyruvate carboxylase* (LOC100749336), and *facilitated trehalose transporter Tret1* (LOC100742930). The broader “catabolic process” GO term was also overrepresented and included multiple upregulated genes associated with autophagy such as *syntaxin-17* (LOC100748805), *autophagy-related 9* (LOC100745297), and *calcium-dependent protein kinase 16-like* (LOC100741566).

Conversely, we observed a downregulation of genes from the 10-min sampling period that function in alternative metabolic pathways. *Phospholipase A1* (LOC100747885), a gene critical in the

breakdown of phospholipids into fatty acids, had a strong relative downregulation ($\log_2$ fold change of -4.28; Table 2), along with other genes relevant in lipid metabolism, including *pancreatic triacylglycerol lipase-like* (LOC100741829), *short-chain specific acyl-CoA dehydrogenase, mitochondrial* (LOC100741796), and *probable Enoyl-CoA hydratase, short chain 1* (LOC100746130). Biological process related GO terms such as “purine nucleobase metabolic process” and “carbohydrate derivative metabolic process” were identified from the downregulated gene list and included DEGs such as *phosphoribosylformylglycinamide synthase* (LOC100749333), *GART trifunctional enzyme* (LOC100745134), and *xanthine dehydrogenase* (LOC100745073; Figure 4; Table 2). Notably, “small molecule metabolic process” was also statistically overrepresented in the downregulated gene set, however, this GO category is broad and also contains upregulated DEGs (Figure 3).

Outside of the scope of metabolic pathway regulation, there were other significant DEGs of interest identified from the upregulated 10-min gene list, including those relating to muscular function [*uncharacterized locus* (LOC100741447, *D. melanogaster* homolog = *myofilin*), *troponin C*, *isoallergen Bla g 6.0201* (LOC10075020), and *troponin T* (LOC100744757)], cytoskeletal restructuring [*PDZ and LIM domain protein 4-like* (LOC100748245), *ankyrin-2* (LOC100744281), and *uncharacterized locus* (LOC100749880, *D. melanogaster* homolog = *supervillin*)], and ion homeostasis [*sodium/potassium-transporting ATPase subunit beta-1-interacting protein* (LOC100744623)]. There was also a mix of up and downregulated DEGs that play a role in the regulation of misfolded proteins during environmental stress [i.e. heat shock proteins (HSPs)] such as *protein lethal (2) essential for life* (LOC100742443; downregulated), *heat shock protein 60A* (LOC100748180; downregulated), *dnaJ homolog subfamily C member 1* (LOC100748155; downregulated), *97 kDa heat shock protein* (LOC100742698; downregulated), and *Chaperone protein DnaJ-like* (LOC105682145; upregulated).

Given the smaller DEG set at 30-min, only a single significantly overrepresented term was recovered, which was related to lipase activity (molecular function: 1-acyl-2-lysophosphatidylserine acylhydrolase activity; Supplemental Material SD.3).

Weighted Gene Co-Expression Network Analysis (WGCNA) Module Results

WGCNA clustered expression data into 20 covarying eigengene expression modules (Table.SI.2). Five of these modules had a significant relationship with a main effect (Table 3; SI.3A-E). In contrast to the clear temporal expression differences observed in the DESeq2 analysis, AIC supported “Treatment Only” as the best-fitting model for all five modules. Nonetheless, this analysis allowed us to identify broader sets of genes and associated terms responding in parallel to the cold treatment without the

need for gene-specific statistical tests. These modules contained many GO terms that overlapped with DEG analyses discussed above, as well as several additional interesting terms (Supplemental Material SD.5). For example, differential gene expression of HSPs were observed in the 10-min sampling period, but GO terms relating to misfolding proteins were much more strongly represented in the WGCNA analyses for two modules; Module 19 was the smallest module (Table 3; Figure 4), with 57 genes that were associated with “endoplasmic reticulum stress”, “protein folding”, “cellular response to topologically incorrect protein”, “chaperone binding”, “misfolded protein binding”, and “unfolded protein binding”. Module 12 (n = 507 genes) was also associated with protein folding regulation (e.g. “protein glycosylation”, “protein localization to the endoplasmic reticulum”; Figure 4) and included other GO terms represented in the 10-min downregulated DEGs, such as “carbohydrate derivative metabolic process” and “catalytic activity”. Module 8 (n= 361 genes) was strongly associated with energy production and mitochondrial function, including GO terms such as “quinone biosynthetic process”, “ATP metabolic process”, and “mitochondrial respiratory chain complex assembly” (Figure 4). The remaining significant modules (modules 1 and 15) represented a wider array of GO terms that include a mix of gene expression regulation and cell signaling terms (Figure 4), including one notable observation for Module 1 (n= 1283): GO terms important for neuromuscular function and synaptic maintenance that are expected to be crucial for chill coma recovery (Kelty et al., 1996). Out of all the modules, Module 1 also contained the most DEGs (n =104) which included top DESeq2 DEGs such as *glycogen phosphorylase* (LOC100746613), *facilitated trehalose transporter Tret1* (LOC100742930), and *sodium/potassium-transporting ATPase subunit beta-1-interacting protein* (LOC100744623).

DISCUSSION

We investigated temporal patterns of gene expression during recovery from an acute chill coma inducing event in the bumble bee *B. impatiens* to test the degree to which the recovery response is immediate and short-lived or might also comprise longer-lived transcriptional modifications that could prime individuals for future cold exposure (Teets et al., 2023). Our results indicate that *B. impatiens* workers undergo a rapid gene regulatory response during recovery from chill coma and after only 30 minutes exhibit no detectable DEGs from bees maintained at ambient temperatures, with most transcriptional changes observed at 10 minutes after removal from cold exposure, shortly after recovery from chill coma (Figure SI.1). Interestingly, no prolonged differential gene expression was detectable at 120-min and 720-min time points. Overall, this suggests that *B. impatiens* workers mount a rapid and

short-lived gene-expression response after cold exposure. These results imply that bumble bees may not cold-harden, at least in relation to resisting and recovering from chill coma, though physiological experiments with additional timing and temperature treatments in conjunction with gene expression and proteomic analyses will be necessary to confirm this. That said, only one other study, to our knowledge, has reported potential cold hardening responses in bumble bees: bees were more likely to survive exposure to -5 °C if they had previously experienced a 1 h exposure to 0 °C or if they were gradually ramped to -5 °C (Owen et al., 2013). Mechanisms providing protection from lethal freezing (Colinet et al., 2018) may differ from those determining CT_{min} and recovery from chill coma (MacMillan et al., 2009; MacMillan & Sinclair, 2011; Overgaard et al., 2005; Teets & Denlinger, 2013; Teets et al., 2019). Further, the short-lived transcriptomic responses documented here may lead to longer lived alterations, for example in membrane properties or metabolite abundances that confer protection against subsequent cold exposure (Colinet et al., 2012; Hazel, 1995). Nevertheless, transcriptomic responses to cold exposure in other (ectothermic) insects often persist for hours to days with clear links to subsequent cold hardening (Overgaard et al., 2007; Teets et al., 2019); heterothermic bumble bees with high metabolic turnover may rely on alternative strategies to persist in highly variable thermal environments.

The identification of GO terms involved in carbohydrate synthesis and metabolism from the upregulated gene set indicates a shift in the metabolic activities of treatment individuals as an integral part in the cold recovery process, consistent with other transcriptomic studies in Hymenoptera (Liu et al., 2020; Xu et al., 2017). One key feature in bumble bee cold tolerance is a capacity to contract thoracic muscles to generate heat (Beenakkers, 1969; Candy et al., 1997; Heinrich, 2004), which is powered by glucose metabolism (Surholt et al., 1991). The gene product of *glycogen phosphorylase* catalyzes glycogenolysis to produce glucose-1-phosphate within muscular tissue (Zhang et al., 2017), while *facilitated trehalose transporter Tret1* is responsible for mobilization of trehalose, the major disaccharide in insect hemolymph that is cleaved to provide glucose for flight and recovery from abiotic stress (Shukla et al., 2015). Trehalose is also a key cryoprotectant important for insect cold hardiness (Arrese & Soulages, 2010; Zeng et al., 2020). In our DEG dataset, upregulation of both genes persisted from 10-min to 30-min, suggesting the importance of carbohydrate utilization in bees recovering from cold exposure. Interestingly, *protein phosphatase 1 regulatory subunit 3C-B*, which stimulates glycogen synthesis and inhibits *glycogen phosphorylase* activity (Agius, 2015), was also upregulated in bees from 10-min but not 30-min. This might suggest that cold exposed bees transition from glycogen synthesis to glycogenolysis within 10 minutes post chill coma recovery, with the catabolism of glycogen for energy

persisting for at least an additional 20 minutes. While other energy reserves such as lipids offer more metabolic energy per molecule compared to carbohydrates (Arrese & Soulages, 2010), glucose in thoracic muscles and circulating in the hemolymph is more easily accessible and faster to metabolize for energy when rapidly recovering from cold exposure. Thus, the presence of downregulated DEGs associated with lipid metabolism [e.g., *Phospholipase A1* (Richmond & Smith, 2011), *Pancreatic triacylglycerol lipase-like* (Lowe, 1997)] was not surprising.

Corroborating the inference that energy derived from glucose plays a major role in the initial muscle activation, *Phosphoenolpyruvate carboxykinase 1* (*Pepck1*), an essential rate-limiting enzyme for gluconeogenesis that helps convert oxaloacetate into phosphoenolpyruvate (PEP; Spacht et al., 2018), was also strongly upregulated at 10-min. The upregulation of *Pepck1* after acute cold exposure is consistent with findings in other insects responding to cold stress and preparing for diapause (Spacht et al., 2018). Other key upregulated genes include *enolase*, which can process PEP into 2-phosphoglycerate in the gluconeogenesis pathway (Melkonian et al., 2019), or conversely serve in the glycolysis pathway by reversing the conversion of 2-phosphoglycerate back to PEP (Avilán et al., 2011), *pyruvate kinase*, that serves as a catalyst for the conversion of PEP into pyruvate (Hsiao et al., 2002; Wiskich, 1980), and *pyruvate carboxylase*, which carboxylates pyruvate to form oxaloacetate in the first step of gluconeogenesis (Melkonian et al., 2019). Module 8 in the WGCNA analysis comprised multiple GO terms associated with mitochondrial activity and ATP synthesis (Figure 4), in which pyruvate plays an important role in the citric acid cycle (Akram 2014). This trend is also observed in other insects that experience both seasonal and rapid cold-hardening (Teets & Denlinger, 2013), suggesting that anaerobic respiration is involved in cold tolerance and subsequent responses across varying timescales (from acute to chronic exposure). Although the above genes do not represent all DEGs observed relating to gluconeogenesis and glycolysis (Table 1; Supplemental Material SD.1), they emphasize the importance of energy derived from glucose for fueling bumble bee thoracic muscles as they recover from chill coma.

Several genes directly associated with muscle and neuromuscular function were detected in our DESeq2 and WGCNA approach, including LOC100741447 (*D. melanogaster* homolog *myofilin*), which is required for striated muscle formation in insects (Qiu et al., 2005), and *Troponin C*, *isoallergen Bla g 6.0201* and *Troponin T*, which facilitates muscle contractions via the binding of calcium (Ca^{2+}) ions, similar to observations in cold-acclimated *Drosophila* (Bayley et al., 2020; Bayley et al., 2018; MacMillan et al., 2016b; Overgaard & MacMillan, 2017). Intracellular $\text{Ca}^{2+}/\text{K}^{+}$ ion homeostasis is vital for the induction and recovery from chill coma (Overgaard & MacMillan, 2017) and genes associated with ion balance have been detected in transcriptomic and genomic scans for selection signatures across

populations exposed to cold temperatures in bumble bees (Jackson et al., 2018; Pimsler et al., 2020). The upregulation of one ion transport gene, *sodium/potassium-transporting ATPase subunit beta-1-interacting protein* (Hilbers et al., 2016), could be indicative of one mechanism of chill coma recovery by repolarizing the membrane potential and restoring neuromuscular function and excitability (Overgaard & MacMillan, 2017). A single gene does not confirm this phenomenon is a major component of the chill coma recovery in our experiment, and differential expression of additional ion transport genes would be required to achieve full muscular repolarization, but the detection is at least consistent with the hypothesis that repolarization may be involved in bees recovering from chill coma. Additionally, expression of the cytoskeletal binding protein encoding *ankryn-2*, suggested to play a role in the stability and modification of *Drosophila* neuromuscular junction synaptic connections (Cunha & Mohler, 2009; Koch et al., 2008), was also significantly increased in our bees recovering from chill coma. Other upregulated genes linked to the cytoskeletal restructuring included *PDZ and LIM domain protein 4-like* and *supervillin* (a well characterized *Drosophila* homolog), which encode for proteins that can alter the actin cytoskeleton and influence cell motility and adhesion (Guryanova et al., 2011; Krcmery et al., 2010; Oh et al., 2003; Smith et al., 2010). Similarly, molecular indicators of cold-induced cytoskeletal reorganization have also been identified in other insect species (Des Marteaux et al., 2017; Des Marteaux et al., 2018; Kim et al., 2006; MacMillan et al., 2016b; Teets et al., 2012), implying an important and possibly conserved nature of this response. From the WGCNA analysis and GO clustering approaches, Module 1 consisted of GO terms such as “cellular signaling”, “synaptic development and regulation”, “locomotion”, and other biological regulators and processes that respond to external stressors (Figure 4; Table SI.3). An increase of eigengene values for this module (treatment relative to control; SD.4) suggests that these functions and processes are actively being restored during the time of rapid recovery, coinciding with the differential expression of associated genes observed in muscular and cytoskeletal genes.

The upregulation of several genes involved in autophagy highlights the role of this process in cold recovery. Autophagy can breakdown glycogen to provide glucose for glycolysis, and also plays an important role in mitigating cellular damage sustained during cold shock (Kaur & Debnath, 2015). Autophagy has been observed as a crucial process in other insects exposed to cold acclimation treatments (Des Marteaux et al., 2017). We also detected differential expression of heat shock proteins (HSPs), although GO analysis only showed significant enrichment in the WGCNA modules. HSPs aid in protein stabilization and refolding after cold exposure (Joplin et al., 1990; King & MacRae, 2015; Lubawy et al., 2022; Rinehart et al., 2007; Vierling, 1991) and have been implicated in winter diapause (King &

MacRae, 2015; Košťál & Tollarová-Borovanská, 2009; Rinehart et al., 2007), cold acclimation (Des Marteaux et al., 2017; MacMillan et al., 2016b; Štětina et al., 2015), and cold shock (Colinet et al., 2010; Lu et al., 2016; Michaud & Denlinger, 2004) for a variety of insect species. *Chaperone protein DnaJ-like* and *dnaJ homolog subfamily C member 1*, both mediators of the Hsp70 family of proteins (Cyr et al., 1994), were differentially expressed in our cold exposed *B. impatiens*, with *DnaJ-like* also identified as a DEG in *B. vosnesenskii* cooled to CT_{min} (Pimsler et al., 2020). Conversely, expression was reduced for *protein lethal (2) essential for life*, *Heat shock protein 60A*, and *97 kDa heat shock protein*, indicating that different HSPs are likely important in different ways during the cold exposure recovery process. Coupled with the decreased eigengene values (treatment relative to control; SD.4) of Module 12, a module comprised of ontologies and terms such as “unfolded protein response”, “glycosylation”, and “glycoprotein”, the differential expression patterns of HSPs may indicate issues with mechanisms regulating protein folding in cold exposed bees. Module 19, which also had reduced eigengene values (SD.4), was overrepresented by GO terms that also relate to protein folding regulation, but more importantly, “response to endoplasmic reticulum stress” (ER stress). As an abundance of unfolded proteins aggregate in the endoplasmic reticulum, ER stress can elicit the unfolded protein response (Ron & Walter, 2007), which can be reversed through programmed cell death to purge afflicted cells. It is possible that our treatment bees experienced ER stress early in their recovery from chill coma, coinciding with the observed HSP expression patterns, the identification of Module 12, and the upregulation of autophagy-related genes.

We have attempted to highlight some of the most important DEGs and gene ontologies that have clear relevance to the chill coma recovery response. However, numerous other processes were detected. For example, we detected the metabolism of purines, which serve as key energy carrier molecules (e.g. ATP and GTP) and play an important role in facilitating proper cell signaling (e.g., cAMP and GMP) (Holland et al., 2011; Ramsey et al., 2010; Tang et al., 2019), and which have also been observed in cold acclimated *Drosophila suzukii* (Enriquez & Colinet, 2019). The differential expression of *xanthine dehydrogenase*, a catalyst for the oxidizing hypoxanthine and xanthine to NADH₂ and uric acid, is particularly intriguing, as similar genes [*xanthine dehydrogenase/oxidase-like* (LOC100741462 and LOC117161180)] have repeatedly been detected in selection scans of bumble bees from different environments, although the function of this gene in *Bombus* adaptation remains elusive (Ghisbain et al., 2020; Heraghty et al., 2022; Pimsler et al., 2017).

CONCLUSION

This study provides insight into transcriptional changes that alter particular biological processes and molecular functions in *B. impatiens* recovering from cold exposure. Notably, differentially expressed genes identified in this study may help elucidate drivers of population variation in thermal tolerance across the geographic ranges of bumble bee species. Many bumble bee species are active early in spring and can fly and forage at colder temperatures than many other insects. Emerging spring gynes (reproductive females), early season workers, and males that have left the colony are especially prone to acute cold events as they venture away from the temperature regulated colony. When bumble bees experience intermittent thermal fluctuations, the ability to thermoregulate, endure cold snaps and rapidly return to biological homeostasis is essential for individual survival. Our study reveals a rapid molecular response following acute cold exposure. High metabolic rates may be sufficient for bumble bees to quickly regain function without the need to maintain long-term transcriptional shifts in anticipation of subsequent cold events. That said, we recognize the possibility that long-term markers of cold hardening may in fact be present, however, they may exist at more subtle log fold changes requiring larger sample sizes per sampling period to enhance detection probability.

For future research, capturing the intermittent transcriptomic profiles of bees recovering from chill coma, particularly, between the 00-min and 10-min interval may be beneficial in characterizing even finer temporal scale recovery responses, prior to, during, and immediately after regaining central nervous system and neuromuscular function. Gene expression differences of alternate tissue types, such as brain and fat body, might also yield interesting results, given their roles in bumble bee sensory and motor function (Mares et al., 2005; Paulk et al., 2008), and energy storage/production (Alford, 1969; Arrese & Soulages, 2010; Costa et al., 2020; Röseler & Röseler, 1986), respectively. An experimental design incorporating individuals across castes (i.e., queens, workers, and males) might also be an interesting research avenue as caste-specific responses may inhibit or enhance function given their respective life history strategies (Goulson 2010). Ultimately, incorporating an integrative multi-omics approach, such as supplementing gene expression data with metabolomic and proteomic datasets and utilization of network-based integration approaches (Zhou et al., 2020), would be ideal to establish connections across multiple levels of biological organization and underlying molecular processes and pathways implicated in chill coma recovery.

ACKNOWLEDGEMENTS

We thank The University of Alabama College of Arts and Sciences and The University of Wyoming College of Agriculture, Life Sciences and Natural Resources. We are also grateful to the University of Texas at Austin Genomic Sequencing and Analysis Facility for library preparation and sequencing.

COMPETING INTERESTS

We declare no competing interests.

AUTHOR CONTRIBUTIONS

KMV processed the samples, performed statistical analysis, and wrote the paper. ECK was responsible for animal rearing, developed the experimental design/ sampling, drafted the animal care/ sampling portion of the Methods section, and data organization/ visualization for the chill coma recovery data. SRR assisted with statistical analysis and editing the paper. MJJ contributed to editing the paper and provided assistance with laboratory work and data interpretation. MED developed the experimental design, provided the facilities and materials for animal husbandry/ experiments, obtained funding, and contributed to editing the paper. JDL developed the experimental design, assisted with sampling processing, obtained funding, helped with statistical analysis, and contributed to editing the paper.

FUNDING

This work was supported by the National Science Foundation (EF-1921585 to JDL, EF-1921562 to MD). Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation.

DATA AVAILABILITY

Raw sequencing data is available on NCBI SRA (Bioproject PRJNA981080; accession numbers SAMN35657356- SAMN35657429). Gene count files and scripts for processing samples are available on FigShare (<https://doi.org/10.6084/m9.figshare.23524047>).

REFERENCES

- Agius, L.** (2015). Role of glycogen phosphorylase in liver glycogen metabolism. *Molecular Aspects of Medicine* **46**, 34-45.
- Alford, D.** (1969). Studies on the fat-body of adult bumble bees. *Journal of Apicultural Research* **8(1)**, 37-48.
- Andersen, J. L., MacMillan, H. A., & Overgaard, J.** (2015). Muscle membrane potential and insect chill coma. *The Journal of Experimental Biology* **218(16)**, 2492-2495.
- Andrews, S.** (2010). Babraham bioinformatics-FastQC a quality control tool for high throughput sequence data. URL: <https://www.bioinformatics.babraham.ac.uk/projects/fastqc>.
- Arrese, E. L., & Soulages, J. L.** (2010). Insect fat body: energy, metabolism, and regulation. *Annual Review of Entomology* **55**, 207-225.
- Avilán, L., Gualdrón-López, M., Quiñones, W., González-González, L., Hannaert, V., Michels, P. A., & Concepción, J.-L.** (2011). Enolase: a key player in the metabolism and a probable virulence factor of trypanosomatid parasites—perspectives for its use as a therapeutic target. *Enzyme Research*, 2011.
- Bale, J.** (2002). Insects and low temperatures: from molecular biology to distributions and abundance. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences* **357(1423)**, 849-862.
- Bale, J. S., Masters, G. J., Hodkinson, I. D., Awmack, C., Bezemer, T. M., Brown, V. K., Butterfield, J., Buse, A., Coulson, J. C., & Farrar, J.** (2002). Herbivory in global climate change research: direct effects of rising temperature on insect herbivores. *Global Change Biology* **8(1)**, 1-16.
- Baniwal, S. K., Bharti, K., Chan, K. Y., Fauth, M., Ganguli, A., Kotak, S., Mishra, S. K., Nover, L., Port, M., & Scharf, K.-D.** (2004). Heat stress response in plants: a complex game with chaperones and more than twenty heat stress transcription factors. *Journal of Biosciences* **29(4)**, 471-487.
- Bates, D., Mächler, M., Bolker, B., & Walker, S.** (2014). Fitting linear mixed-effects models using lme4. *arXiv Preprint arXiv:1406.5823*.

- Bayley, J. S., Sørensen, J. G., Moos, M., Košťál, V., & Overgaard, J.** (2020). Cold acclimation increases depolarization resistance and tolerance in muscle fibers from a chill-susceptible insect, *Locusta migratoria*. *American Journal of Physiology-Regulatory, Integrative and Comparative Physiology* **319**(4), R439-R447.
- Bayley, J. S., Winther, C. B., Andersen, M. K., Grønkjær, C., Nielsen, O. B., Pedersen, T. H., & Overgaard, J.** (2018). Cold exposure causes cell death by depolarization-mediated Ca²⁺ overload in a chill-susceptible insect. *Proceedings of the National Academy of Sciences* **115**(41), E9737-E9744.
- Beenackers, A. T.** (1969). Carbohydrate and fat as a fuel for insect flight. A comparative study. *Journal of Insect Physiology* **15**(3), 353-361.
- Benjamini, Y., & Hochberg, Y.** (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)* **57**(1), 289-300.
- Bowler, K., & Terblanche, J. S.** (2008). Insect thermal tolerance: what is the role of ontogeny, ageing and senescence? *Biological Reviews* **83**(3), 339-355.
- Buckley, B. A., & Somero, G. N.** (2009). cDNA microarray analysis reveals the capacity of the cold-adapted Antarctic fish *Trematomus bernacchii* to alter gene expression in response to heat stress. *Polar Biology* **32**, 403-415.
- Candy, D., Becker, A., & Wegener, G.** (1997). Coordination and integration of metabolism in insect flight. *Comparative Biochemistry and Physiology Part B: Biochemistry and Molecular Biology* **117**(4), 497-512.
- Chen, H., & Boutros, P. C.** (2011). VennDiagram: a package for the generation of highly-customizable Venn and Euler diagrams in R. *BMC bioinformatics* **12**(1), 1-7.
- Colinet, H., Larvor, V., Laparie, M., & Renault, D.** (2012). Exploring the plastic response to cold acclimation through metabolomics. *Functional Ecology* **26**(3), 711-722.
- Colinet, H., Lee, S. F., & Hoffmann, A.** (2010). Temporal expression of heat shock genes during cold stress and recovery from chill coma in adult *Drosophila melanogaster*. *The FEBS Journal* **277**(1), 174-185.

- Colinet, H., Rinehart, J. P., Yocum, G. D., & Greenlee, K. J.** (2018). Mechanisms underpinning the beneficial effects of fluctuating thermal regimes in insect cold tolerance. *Journal of Experimental Biology* **221**(14), jeb164806.
- Colinet, H., Sinclair, B. J., Vernon, P., & Renault, D.** (2015). Insects in fluctuating thermal environments. *Annual Review of Entomology* **60**, 123.
- Costa, C. P., Duennes, M. A., Fisher, K., Der, J. P., Watrous, K. M., Okamoto, N., Yamanaka, N., & Woodard, S. H.** (2020). Transcriptome analysis reveals nutrition-and age-related patterns of gene expression in the fat body of pre-overwintering bumble bee queens. *Molecular Ecology* **29**(4), 720-737.
- Cunha, S. R., & Mohler, P. J.** (2009). Ankyrin protein networks in membrane formation and stabilization. *Journal of Cellular and Molecular Medicine* **13**(11-12), 4364-4376.
- Cyr, D. M., Langer, T., & Douglas, M. G.** (1994). DnaJ-like proteins: molecular chaperones and specific regulators of Hsp70. *Trends in Biochemical Sciences* **19**(4), 176-181.
- Denlinger, D. L., Rinehart, J. P., & Yocum, G. D.** (2001). Stress proteins: a role in insect diapause? *Insect Timing: Circadian Rhythmicity to Seasonality*, 155-171.
- Des Marteaux, L. E., McKinnon, A. H., Udaka, H., Toxopeus, J., & Sinclair, B. J.** (2017). Effects of cold-acclimation on gene expression in Fall field cricket (*Gryllus pennsylvanicus*) ionoregulatory tissues. *BMC Genomics* **18**(1), 1-17.
- Des Marteaux, L. E., Stinziano, J. R., & Sinclair, B. J.** (2018). Effects of cold acclimation on rectal macromorphology, ultrastructure, and cytoskeletal stability in *Gryllus pennsylvanicus* crickets. *Journal of Insect Physiology* **104**, 15-24.
- DeSalvo, M., Voolstra, C. R., Sunagawa, S., Schwarz, J., Stillman, J., Coffroth, M. A., Szmant, A., & Medina, M.** (2008). Differential gene expression during thermal stress and bleaching in the Caribbean coral *Montastraea faveolata*. *Molecular ecology* **17**(17), 3952-3971.
- Dobin, A., Davis, C. A., Schlesinger, F., Drenkow, J., Zaleski, C., Jha, S., Batut, P., Chaisson, M., & Gingeras, T. R.** (2013). STAR: ultrafast universal RNA-seq aligner. *Bioinformatics* **29**(1), 15-21.

- Enriquez, T., & Colinet, H.** (2019). Cold acclimation triggers lipidomic and metabolic adjustments in the spotted wing drosophila *Drosophila suzukii* (Matsumura). *American Journal of Physiology-Regulatory, Integrative and Comparative Physiology* **316**(6), R751-R763.
- Esch, H., Goller, F., & Heinrich, B.** (1991). How do bees shiver? *The Science of Nature*, 78(7), 325-328.
- Free, J.** (1955). The collection of food by bumblebees. *Insectes Sociaux* **2**, 303-311.
- Ghisbain, G., Lozier, J. D., Rahman, S. R., Ezray, B. D., Tian, L., Ulmer, J. M., Heraghty, S. D., Strange, J. P., Rasmont, P., & Hines, H. M.** (2020). Substantial genetic divergence and lack of recent gene flow support cryptic speciation in a colour polymorphic bumble bee (*Bombus bifarius*) species complex. *Systematic Entomology* **45**(3), 635-652.
- Goulson, D.** (2010). *Bumblebees: behaviour, ecology, and conservation*. Oxford University Press on Demand.
- Grote, S.** (2018). GOfuncR: Gene ontology enrichment using FUNC. *R Package Version*, 1(0).
- Gu, Z., & Hübschmann, D.** (2022). Simplify enrichment: A bioconductor package for clustering and visualizing functional enrichment results. *Genomics, Proteomics & Bioinformatics*.
- Guryanova, O. A., Drazba, J. A., Frolova, E. I., & Chumakov, P. M.** (2011). Actin cytoskeleton remodeling by the alternatively spliced isoform of PDLIM4/RIL protein. *Journal of Biological Chemistry* **286**(30), 26849-26859.
- Harvey, J. A., Tougeron, K., Gols, R., Heinen, R., Abarca, M., Abram, P. K., Basset, Y., Berg, M., Boggs, C., & Brodeur, J.** (2023). Scientists' warning on climate change and insects. *Ecological Monographs* **93**(1), e1553.
- Hazel, J. R.** (1995). Thermal adaptation in biological membranes: is homeoviscous adaptation the explanation? *Annual Review of Physiology* **57**(1), 19-42.
- Heikkilä, J. J.** (2017). The expression and function of hsp30-like small heat shock protein genes in amphibians, birds, fish, and reptiles. *Comparative Biochemistry and Physiology Part A: Molecular & Integrative Physiology* **203**, 179-192.
- Heinrich, B.** (1993). Bumblebees out in the cold. In *The Hot-Blooded Insects. Strategies and Mechanisms of Thermoregulation*, 227-276. Springer.

- Heinrich, B.** (2004). *Bumblebee economics*. Harvard University Press.
- Heinrich, B., & Esch, H.** (1994). Thermoregulation in bees. *American Scientist* **82**(2), 164-170.
- Heraghty, S. D., Rahman, S. R., Jackson, J. M., & Lozier, J. D.** (2022). Whole Genome Sequencing Reveals the Structure of Environment-Associated Divergence in a Broadly Distributed Montane Bumble Bee, *Bombus vancouverensis*. *Insect Systematics and Diversity* **6**(5), 5.
- Hilbers, F., Kopec, W., Isaksen, T. J., Holm, T. H., Lykke-Hartmann, K., Nissen, P., Khandelia, H., & Poulsen, H.** (2016). Tuning of the Na, K-ATPase by the beta subunit. *Scientific Reports* **6**(1), 1-11.
- Hines, H. M.** (2008). Historical biogeography, divergence times, and diversification patterns of bumble bees (Hymenoptera: Apidae: Bombus). *Systematic Biology* **57**(1), 58-75.
- Holland, C., Lipsett, D. B., & Clark, D. V.** (2011). A link between impaired purine nucleotide synthesis and apoptosis in *Drosophila melanogaster*. *Genetics* **188**(2), 359-367.
- Hsiao, P. F., Zhu, Y.-J., & Chien, Y.-C.** (2002). Cloning and functional analysis of pyruvate kinase promoter region from *Drosophila melanogaster*. *DNA and Cell Biology* **21**(1), 1-10.
- Jackson, J. M., Pimsler, M. L., Oyen, K. J., Koch-Uhuad, J. B., Herndon, J. D., Strange, J. P., Dillon, M. E., & Lozier, J. D.** (2018). Distance, elevation and environment as drivers of diversity and divergence in bumble bees across latitude and altitude. *Molecular Ecology* **27**(14), 2926-2942.
- Joplin, K. H., Yocum, G. D., & Denlinger, D. L.** (1990). Cold shock elicits expression of heat shock proteins in the flesh fly, *Sarcophaga crassipalpis*. *Journal of Insect Physiology* **36**(11), 825-834.
- Kaur, J., & Debnath, J.** (2015). Autophagy at the crossroads of catabolism and anabolism. *Nature Reviews Molecular Cell Biology* **16**(8), 461-472.
- Keaveny, E. C., Waybright, S. A., Rusch, T. W., & Dillon, M. E.** (2022). Supercooling points of freeze-avoiding bumble bees vary with caste and queen life stage. *Journal of Thermal Biology* **104**, 103196.

- Kellermann, V., Overgaard, J., Hoffmann, A. A., Fløjgaard, C., Svenning, J.-C., & Loeschcke, V. (2012). Upper thermal limits of *Drosophila* are linked to species distributions and strongly constrained phylogenetically. *Proceedings of the National Academy of Sciences* **109**(40), 16228-16233.
- Kelty, J. D., Killian, K. A., & Lee, R. E. (1996). Cold shock and rapid cold-hardening of pharate adult flesh flies (*Sarcophaga crassipalpis*): effects on behaviour and neuromuscular function following eclosion. *Physiological Entomology* **21**(4), 283-288.
- Kim, M., Robich, R. M., Rinehart, J. P., & Denlinger, D. L. (2006). Upregulation of two actin genes and redistribution of actin during diapause and cold stress in the northern house mosquito, *Culex pipiens*. *Journal of Insect Physiology* **52**(11-12), 1226-1233.
- King, A. M., & MacRae, T. H. (2015). Insect heat shock proteins during stress and diapause. *Annual Review of Entomology* **60**, 59-75.
- Kingsolver, J. G., Arthur Woods, H., Buckley, L. B., Potter, K. A., MacLean, H. J., & Higgins, J. K. (2011). Complex life cycles and the responses of insects to climate change. In: Oxford University Press.
- Koch, I., Schwarz, H., Beuchle, D., Goellner, B., Langeegger, M., & Aberle, H. (2008). *Drosophila* ankyrin 2 is required for synaptic stability. *Neuron* **58**(2), 210-222.
- Košťál, V., & Tollarova-Borovanska, M. (2009). The 70 kDa heat shock protein assists during the repair of chilling injury in the insect, *Pyrrhocoris apterus*. *PLoS One* **4**(2), e4546.
- Krcmery, J., Camarata, T., Kulisz, A., & Simon, H. G. (2010). Nucleocytoplasmic functions of the PDZ-LIM protein family: new insights into organ development. *Bioessays* **32**(2), 100-108.
- Kuznetsova, A., Brockhoff, P. B., & Christensen, R. H. (2017). lmerTest package: tests in linear mixed effects models. *Journal of Statistical Software* **82**, 1-26.
- Lande, R. (2009). Adaptation to an extraordinary environment by evolution of phenotypic plasticity and genetic assimilation. *Journal of Evolutionary Biology* **22**(7), 1435-1446.
- Langfelder, P., & Horvath, S. (2008). WGCNA: an R package for weighted correlation network analysis. *BMC Bioinformatics* **9**(1), 1-13.

- Liu, Y., Jin, H., Naeem, M., & An, J.** (2020). Comparative transcriptome analysis reveals regulatory genes involved in cold tolerance and hypoxic adaptation of high-altitude Tibetan bumblebees. *Apidologie* **51**(6), 1166-1181.
- Lohman, B. K., Weber, J. N., & Bolnick, D. I.** (2016). Evaluation of TagSeq, a reliable low-cost alternative for RNA seq. *Molecular Ecology Resources* **16**(6), 1315-1321.
- Love, M. I., Huber, W., & Anders, S.** (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology* **15**(12), 1-21.
- Lowe, M. E.** (1997). Molecular mechanisms of rat and human pancreatic triglyceride lipases. *The Journal of Nutrition* **127**(4), 549-557.
- Lu, M.-X., Li, H.-B., Zheng, Y.-T., Shi, L., & Du, Y.-Z.** (2016). Identification, genomic organization and expression profiles of four heat shock protein genes in the western flower thrips, *Frankliniella occidentalis*. *Journal of Thermal Biology* **57**, 110-118.
- Lubawy, J., Chowański, S., Adamski, Z., & Słocińska, M.** (2022). Mitochondria as a target and central hub of energy division during cold stress in insects. *Frontiers in Zoology* **19**(1), 1-19.
- MacMillan, H. A., Findsen, A., Pedersen, T. H., & Overgaard, J.** (2014). Cold-induced depolarization of insect muscle: differing roles of extracellular K⁺ during acute and chronic chilling. *Journal of Experimental Biology* **217**(16), 2930-2938.
- MacMillan, H. A., Guglielmo, C. G., & Sinclair, B. J.** (2009). Membrane remodeling and glucose in *Drosophila melanogaster*: a test of rapid cold-hardening and chilling tolerance hypotheses. *Journal of Insect Physiology* **55**(3), 243-249.
- MacMillan, H. A., Knee, J. M., Dennis, A. B., Udaka, H., Marshall, K. E., Merritt, T. J., & Sinclair, B. J.** (2016a). Cold acclimation wholly reorganizes the *Drosophila melanogaster* transcriptome and metabolome. *Scientific Reports* **6**(1), 28999.
- MacMillan, H. A., Knee, J. M., Dennis, A. B., Udaka, H., Marshall, K. E., Merritt, T. J., & Sinclair, B. J.** (2016b). Cold acclimation wholly reorganizes the *Drosophila melanogaster* transcriptome and metabolome. *Scientific Reports* **6**(1), 1-14.
- MacMillan, H. A., & Sinclair, B. J.** (2011). Mechanisms underlying insect chill-coma. *Journal of Insect Physiology* **57**(1), 12-20.

- MacRae, T. H.** (2010). Gene expression, metabolic regulation and stress tolerance during diapause. *Cellular and Molecular Life Sciences* **67**(14), 2405-2424.
- Maebe, K., De Baets, A., Vandamme, P., Vereecken, N. J., Michez, D., & Smagghe, G.** (2021). Impact of intraspecific variation on measurements of thermal tolerance in bumble bees. *Journal of Thermal Biology* **99**, 103002.
- Maebe, K., Hart, A. F., Marshall, L., Vandamme, P., Vereecken, N. J., Michez, D., & Smagghe, G.** (2021). Bumblebee resilience to climate change, through plastic and adaptive responses. *Global Change Biology* **27**(18), 4223-4237.
- Makinson, J. C., Woodgate, J. L., Reynolds, A., Capaldi, E. A., Perry, C. J., & Chittka, L.** (2019). Harmonic radar tracking reveals random dispersal pattern of bumblebee (*Bombus terrestris*) queens after hibernation. *Scientific Reports* **9**(1), 4651.
- Malmendal, A., Overgaard, J., Bundy, J. G., Sørensen, J. G., Nielsen, N. C., Loeschcke, V., & Holmstrup, M.** (2006). Metabolomic profiling of heat stress: hardening and recovery of homeostasis in *Drosophila*. *American Journal of Physiology-Regulatory, Integrative and Comparative Physiology* **291**(1), R205-R212.
- Mares, S., Ash, L., & Gronenberg, W.** (2005). Brain allometry in bumblebee and honey bee workers. *Brain, Behavior and Evolution* **66**(1), 50-61.
- Martin, M.** (2011). Cutadapt removes adapter sequences from high-throughput sequencing reads. *EMBnet. Journal* **17**(1), 10-12.
- Martinet, B., Lecocq, T., Brasero, N., Biella, P., Urbanová, K., Valterová, I., Cornalba, M., Gjershaug, J. O., Michez, D., & Rasmont, P.** (2018). Following the cold: geographical differentiation between interglacial refugia and speciation in the arcto-alpine species complex *Bombus monticola* (Hymenoptera: Apidae). *Systematic Entomology* **43**(1), 200-217.
- Melkonian, E. A., Asuka, E., & Schury, M. P.** (2019). Physiology, gluconeogenesis. In *StatPearls [Internet]*. StatPearls Publishing
- Meyer, E., Aglyamova, G., & Matz, M.** (2011). Profiling gene expression responses of coral larvae (*Acropora millepora*) to elevated temperature and settlement inducers using a novel RNA-Seq procedure. *Molecular Ecology* **20**(17), 3599-3616.

- Michaud, M. R., & Denlinger, D.** (2004). Molecular modalities of insect cold survival: current understanding and future trends. *International Congress Series* **1275**, 32-46. Elsevier
- Müller, C. B., & Schmid-Hempel, P.** (1993). Exploitation of cold temperature as defence against parasitoids in bumblebees. *Nature* **363(6424)**, 65-67.
- Neven, L. G.** (2000). Physiological responses of insects to heat. *Postharvest Biology and Technology* **21(1)**, 103-111.
- Oh, S. W., Pope, R. K., Smith, K. P., Crowley, J. L., Nebl, T., Lawrence, J. B., & Luna, E. J.** (2003). Archvillin, a muscle-specific isoform of supervillin, is an early expressed component of the costameric membrane skeleton. *Journal of Cell Science* **116(11)**, 2261-2275.
- Overgaard, J., & MacMillan, H. A.** (2017). The integrative physiology of insect chill tolerance. *Annual Review of Physiology* **79**, 187-208.
- Overgaard, J., Malmendal, A., Sørensen, J. G., Bundy, J. G., Loeschcke, V., Nielsen, N. C., & Holmstrup, M.** (2007). Metabolomic profiling of rapid cold hardening and cold shock in *Drosophila melanogaster*. *Journal of Insect Physiology* **53(12)**, 1218-1232.
- Overgaard, J., Sørensen, J. G., Petersen, S. O., Loeschcke, V., & Holmstrup, M.** (2005). Changes in membrane lipid composition following rapid cold hardening in *Drosophila melanogaster*. *Journal of Insect Physiology* **51(11)**, 1173-1182.
- Owen, E. L., Bale, J. S., & Hayward, S. A.** (2013). Can winter-active bumblebees survive the cold? Assessing the cold tolerance of *Bombus terrestris audax* and the effects of pollen feeding. *PLoS One* **8(11)**, e80061.
- Oyen, K. J., Giri, S., & Dillon, M. E.** (2016). Altitudinal variation in bumble bee (*Bombus*) critical thermal limits. *Journal of Thermal Biology* **59**, 52-57.
- Paulk, A. C., Phillips-Portillo, J., Dacks, A. M., Fellous, J.-M., & Gronenberg, W.** (2008). The processing of color, motion, and stimulus timing are anatomically segregated in the bumblebee brain. *Journal of Neuroscience* **28(25)**, 6319-6332.
- Peters, M. K., Peisker, J., Steffan-Dewenter, I., & Hoiss, B.** (2016). Morphological traits are linked to the cold performance and distribution of bees along elevational gradients. *Journal of Biogeography* **43(10)**, 2040-2049.

- Pimsler, M. L., Jackson, J. M., & Lozier, J. D.** (2017). Population genomics reveals a candidate gene involved in bumble bee pigmentation. *Ecology and Evolution* **7**(10), 3406-3413.
- Pimsler, M. L., Oyen, K. J., Herndon, J. D., Jackson, J. M., Strange, J. P., Dillon, M. E., & Lozier, J. D.** (2020). Biogeographic parallels in thermal tolerance and gene expression variation under temperature stress in a widespread bumble bee. *Scientific Reports* **10**(1), 1-11.
- Qiu, F., Brendel, S., Cunha, P. M., Astola, N., Song, B., Furlong, E. E., Leonard, K. R., & Bullard, B.** (2005). Myofilin, a protein in the thick filaments of insect muscle. *Journal of Cell Science* **118**(7), 1527-1536.
- Ramsey, J., MacDonald, S., Jander, G., Nakabachi, A., Thomas, G., & Douglas, A.** (2010). Genomic evidence for complementary purine metabolism in the pea aphid, *Acyrtosiphon pisum*, and its symbiotic bacterium *Buchnera aphidicola*. *Insect Molecular Biology* **19**, 241-248.
- Richmond, G. S., & Smith, T. K.** (2011). Phospholipases A1. *International Journal of Molecular Sciences* **12**(1), 588-612.
- Rinehart, J. P., Li, A., Yocum, G. D., Robich, R. M., Hayward, S. A., & Denlinger, D. L.** (2007). Up-regulation of heat shock proteins is essential for cold survival during insect diapause. *Proceedings of the National Academy of Sciences* **104**(27), 11130-11137.
- Robertson, R. M., Spong, K. E., & Srithiphaphirom, P.** (2017). Chill coma in the locust, *Locusta migratoria*, is initiated by spreading depolarization in the central nervous system. *Scientific Reports* **7**(1), 1-12.
- Ron, D., & Walter, P.** (2007). Signal integration in the endoplasmic reticulum unfolded protein response. *Nature Reviews Molecular Cell Biology* **8**(7), 519-529.
- Röseler, P.-F., & Röseler, I.** (1986). Caste specific differences in fat body glycogen metabolism of the bumblebee, *Bombus terrestris*. *Insect Biochemistry* **16**(3), 501-508.
- Sadd, B. M., Barribeau, S. M., Bloch, G., De Graaf, D. C., Dearden, P., Elsik, C. G., Gadau, J., Grimmelikhuijzen, C. J., Hasselmann, M., & Lozier, J. D.** (2015). The genomes of two key bumblebee species with primitive eusocial organization. *Genome Biology* **16**(1), 1-32.
- Schlichting, C. D., & Pigliucci, M.** (1993). Control of phenotypic plasticity via regulatory genes. *The American Naturalist* **142**(2), 366-370.

- Schoville, S. D., Barreto, F. S., Moy, G. W., Wolff, A., & Burton, R. S.** (2012). Investigating the molecular basis of local adaptation to thermal stress: population differences in gene expression across the transcriptome of the copepod *Tigriopus californicus*. *BMC Evolutionary Biology* **12**(1), 1-17.
- Shearer, P. W., West, J. D., Walton, V. M., Brown, P. H., Svetec, N., & Chiu, J. C.** (2016). Seasonal cues induce phenotypic plasticity of *Drosophila suzukii* to enhance winter survival. *BMC Ecology* **16**(1), 1-18.
- Shen, L. J. R. P.** (2014). GeneOverlap: An R package to test and visualize gene overlaps. *R Package*, 3.
- Shukla, E., Thorat, L. J., Nath, B. B., & Gaikwad, S. M.** (2015). Insect trehalase: physiological significance and potential applications. *Glycobiology* **25**(4), 357-367.
- Sinclair, B., Gibbs, A., & Roberts, S.** (2007). Gene transcription during exposure to, and recovery from, cold and desiccation stress in *Drosophila melanogaster*. *Insect Molecular Biology* **16**(4), 435-443.
- Sinclair, B. J.** (2001). Field ecology of freeze tolerance: interannual variation in cooling rates, freeze-thaw and thermal stress in the microhabitat of the alpine cockroach *Celatoblatta quinque maculata*. *Oikos* **93**(2), 286-293.
- Smith, T. C., Fang, Z., & Luna, E. J.** (2010). Novel interactors and a role for supervillin in early cytokinesis. *Cytoskeleton* **67**(6), 346-364.
- Sonna, L. A., Fujita, J., Gaffin, S. L., & Lilly, C. M.** (2002). Invited review: effects of heat and cold stress on mammalian gene expression. *Journal of Applied Physiology* **92**(4), 1725-1742.
- Spacht, D. E., Teets, N. M., & Denlinger, D. L.** (2018). Two isoforms of Pepck in *Sarcophaga bullata* and their distinct expression profiles through development, diapause, and in response to stresses of cold and starvation. *Journal of Insect Physiology* **111**, 41-46.
- Stange, E. E., & Ayres, M. P.** (2010). Climate change impacts: Insects. *eLS*.
- Štětina, T., Košťál, V., & Korbelová, J.** (2015). The role of inducible Hsp70, and other heat shock proteins, in adaptive complex of cold tolerance of the fruit fly (*Drosophila melanogaster*). *PLoS One* **10**(6), e0128976.

- Stomp, M., van Dijk, M. A., van Overzee, H. M., Wortel, M. T., Sigon, C. A., Egas, M., Hoogveld, H., Gons, H. J., & Huisman, J.** (2008). The timescale of phenotypic plasticity and its impact on competition in fluctuating environments. *The American Naturalist* **172**(5), E169-E185.
- Sunday, J. M., Bates, A. E., & Dulvy, N. K.** (2011). Global analysis of thermal tolerance and latitude in ectotherms. *Proceedings of the Royal Society B: Biological Sciences* **278**(1713), 1823-1830.
- Sunday, J. M., Bates, A. E., & Dulvy, N. K.** (2012). Thermal tolerance and the global redistribution of animals. *Nature Climate Change* **2**(9), 686-690.
- Sunday, J. M., Bates, A. E., Kearney, M. R., Colwell, R. K., Dulvy, N. K., Longino, J. T., & Huey, R. B.** (2014). Thermal-safety margins and the necessity of thermoregulatory behavior across latitude and elevation. *Proceedings of the National Academy of Sciences* **111**(15), 5610-5615.
- Supek, F., Bošnjak, M., Škunca, N., & Šmuc, T.** (2011). REVIGO summarizes and visualizes long lists of gene ontology terms. *PLoS One* **6**(7), e21800.
- Surholt, B., Greive, H., Baal, T., & Bertsch, A.** (1991). Warm-up and substrate cycling in flight muscles of male bumblebees, *Bombus terrestris*. *Comparative Biochemistry and Physiology Part A: Physiology* **98**(2), 299-303.
- Tang, Z., Ye, W., Chen, H., Kuang, X., Guo, J., Xiang, M., Peng, C., Chen, X., & Liu, H.** (2019). Role of purines in regulation of metabolic reprogramming. *Purinergic Signalling* **15**(4), 423-438.
- Teets, N. M., & Denlinger, D. L.** (2013). Physiological mechanisms of seasonal and rapid cold-hardening in insects. *Physiological Entomology* **38**(2), 105-116.
- Teets, N. M., Gantz, J., & Kawarasaki, Y.** (2020). Rapid cold hardening: ecological relevance, physiological mechanisms and new perspectives. *Journal of Experimental Biology* **223**(3), jeb203448.
- Teets, N. M., Kawarasaki, Y., Potts, L. J., Philip, B. N., Gantz, J., Denlinger, D. L., & Lee Jr, R. E.** (2019). Rapid cold hardening protects against sublethal freezing injury in an Antarctic insect. *Journal of Experimental Biology* **222**(15), jeb206011.

- Teets, N. M., Marshall, K. E., & Reynolds, J. A.** (2023). Molecular Mechanisms of Winter Survival. *Annual Review of Entomology* **68**, 319-339.
- Teets, N. M., Peyton, J. T., Ragland, G. J., Colinet, H., Renault, D., Hahn, D. A., & Denlinger, D. L.** (2012). Combined transcriptomic and metabolomic approach uncovers molecular mechanisms of cold tolerance in a temperate flesh fly. *Physiological Genomics* **44(15)**, 764-777.
- Ummenhofer, C. C., & Meehl, G. A.** (2017). Extreme weather and climate events with ecological relevance: a review. *Philosophical Transactions of the Royal Society B: Biological Sciences* **372(1723)**, 20160135.
- Vierling, E.** (1991). The roles of heat shock proteins in plants. *Annual Review of Plant Biology* **42(1)**, 579-620.
- Walsh, A. T., Triant, D. A., Le Tourneau, J. J., Shamimuzzaman, M., & Elsik, C. G.** (2022). Hymenoptera Genome Database: new genomes and annotation datasets for improved go enrichment and orthologue analyses. *Nucleic Acids Research* **50(D1)**, D1032-D1039.
- Walter, W., Sánchez-Cabo, F., & Ricote, M.** (2015). GOplot: an R package for visually combining expression data with functional analysis. *Bioinformatics* **31(17)**, 2912-2914.
- Walther-Hellwig, K., & Frankl, R.** (2000). Foraging habitats and foraging distances of bumblebees, *Bombus* spp.(Hym., Apidae), in an agricultural landscape. *Journal of Applied Entomology* **124(7-8)**, 299-306.
- Wang, X.-h., & Kang, L.** (2003). Rapid cold hardening in young hoppers of the migratory locust *Locusta migratoria* L.(Orthoptera: Acridiidae). *CryoLetters* **24(5)**, 331-340.
- West-Eberhard, M. J.** (1989). Phenotypic plasticity and the origins of diversity. *Annual Review of Ecology and Systematics* **20(1)**, 249-278.
- Whitman, D. W., & Agrawal, A. A.** (2009). What is phenotypic plasticity and why is it important. *Phenotypic Plasticity of Insects: Mechanisms and Consequences*, 1-63.
- Wickham, H.** (2011). ggplot2. *Wiley Interdisciplinary Reviews: Computational Statistics* **3(2)**, 180-185.

- Williams, P. H., Lobo, J. M., & Meseguer, A. S.** (2018). Bumblebees take the high road: climatically integrative biogeography shows that escape from Tibet, not Tibetan uplift, is associated with divergences of present-day *Mendacibombus*. *Ecography* **41**(3), 461-477.
- Williams, P. H., Thorp, R. W., Richardson, L. L., & Colla, S. R.** (2014). Bumble Bees of North America. In *Bumble Bees of North America*. Princeton University Press.
- Wiskich, J.** (1980). Control of the Krebs cycle. In *Metabolism and Respiration*, 243-278. Elsevier.
- Xu, K., Niu, Q., Zhao, H., Du, Y., & Jiang, Y.** (2017). Transcriptomic analysis to uncover genes affecting cold resistance in the Chinese honey bee (*Apis cerana cerana*). *PloS One* **12**(6), e0179922.
- Zeng, B., Wang, S., Li, Y., Xiao, Z., Zhou, M., Wang, S., & Zhang, D.** (2020). Effect of long-term cold storage on trehalose metabolism of pre-wintering *Harmonia axyridis* adults and changes in morphological diversity before and after wintering. *PLoS One* **15**(3), e0230435.
- Zhang, J., Marshall, K. E., Westwood, J. T., Clark, M. S., & Sinclair, B. J.** (2011). Divergent transcriptomic responses to repeated and single cold exposures in *Drosophila melanogaster*. *Journal of Experimental Biology* **214**(23), 4021-4029.
- Zhang, L., Wang, H., Chen, J., Shen, Q., Wang, S., Xu, H., & Tang, B.** (2017). Glycogen phosphorylase and glycogen synthase: gene cloning and expression analysis reveal their role in trehalose metabolism in the Brown planthopper, *Nilaparvata lugens* Stål (Hemiptera: Delphacidae). *Journal of Insect Science* **17**(2), 42.
- Zhao, L., & Jones, W.** (2012). Expression of heat shock protein genes in insect stress responses. *Invertebrate Survival Journal* **9**(1), 93-101.
- Zhou, G., Li, S., & Xia, J.** (2020). Network-based approaches for multi-omics integration. *Computational Methods and Data Analysis for Metabolomics*, 469-487.

Figures and Tables

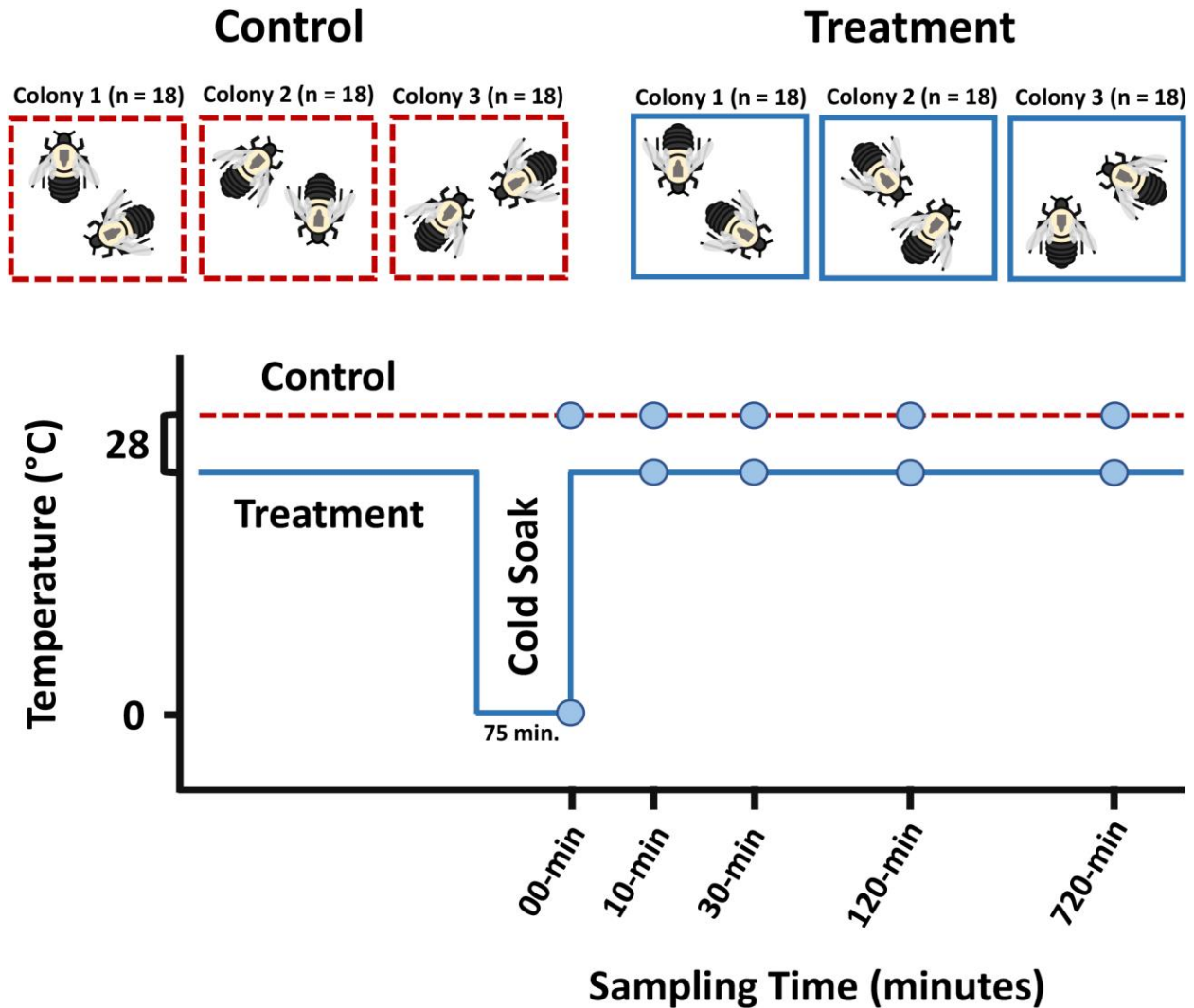


Fig. 1. Experimental design for the acute cold exposure. The six microcolonies (n=18 workers) were divided between treatment (submerged in ice water for 75 minutes, then placed at 28°C for remaining time) and control groups (kept at 28°C). Circles on the time graph indicate sampling times, where three bees were randomly selected from each control and treatment microcolony (N=9 per treatment total).

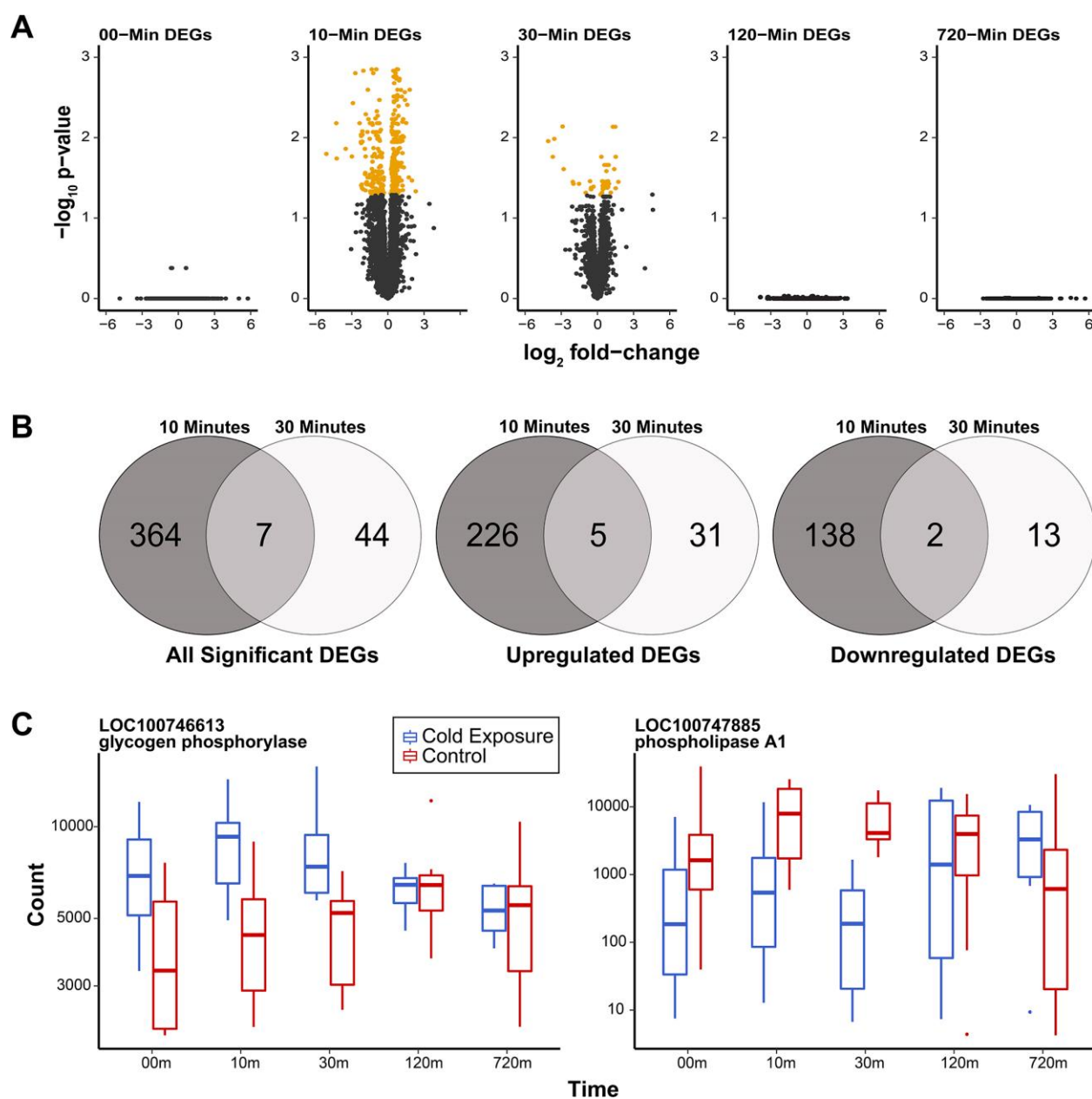


Fig. 2. Results from the differential gene expression analysis. (A) Volcano plot graphing the DEGs (yellow points are significant DEGs, black points are non-significant) in terms of statistical significance (adjusted p-value) versus the magnitude of expression change ($\log_2$ FoldChange). Positive and negative $\log_2$ FoldChange values represent genes up- or down-regulated in response to cold exposure, respectively. (B) Venn diagrams show the significant (adjusted p-value ≤ 0.05) differentially expressed genes for 10-min and 30-min sampling periods. Results from Fisher's exact test indicated that DEG overlaps are statistically greater than expected by

chance (p-values = 0.00059, 0.00042, and 0.01). Although statistically significant, there should be cautious interpretations of comparisons with few DEGs and overlaps between time series.

(C) Boxplots show the gene counts of the treatment (blue) and control (red) bees for two example genes highlighting differential up and downregulation (glycogen phosphorylase and phospholipase A1, respectively) at both the 10-min and 30-min sampling period. A temporal pattern is observed in both plots showing noticeable gene count differences at 10, 30, and even 00 minutes (although it was not statistically significant for this time point).

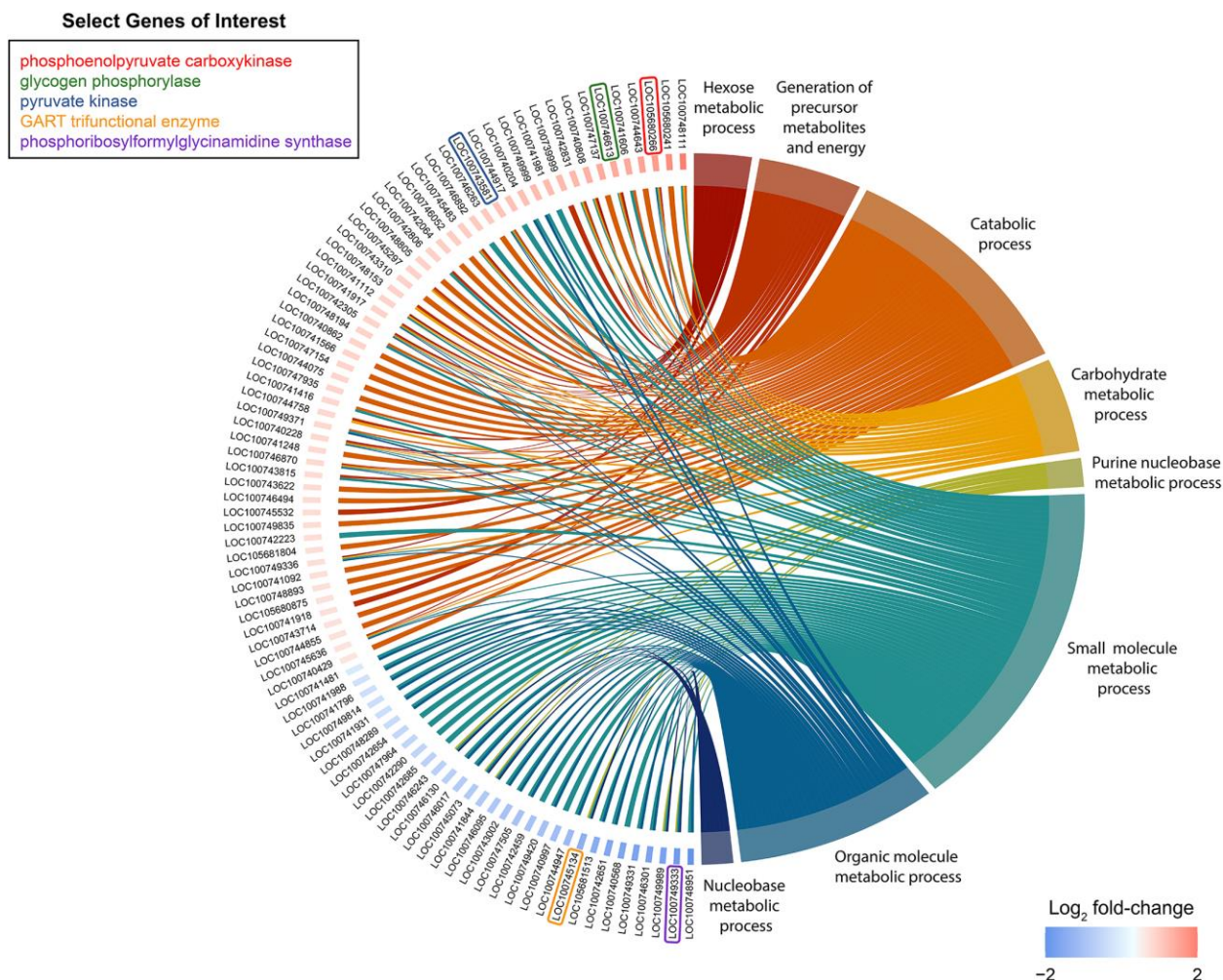
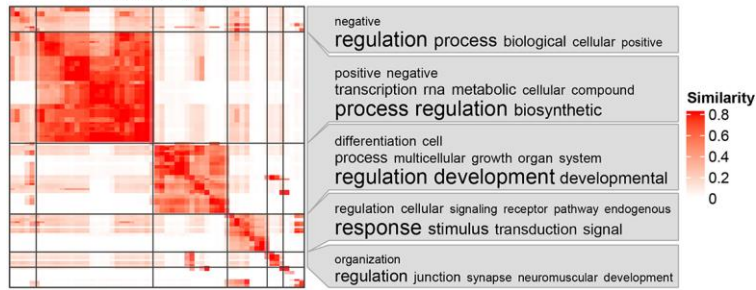
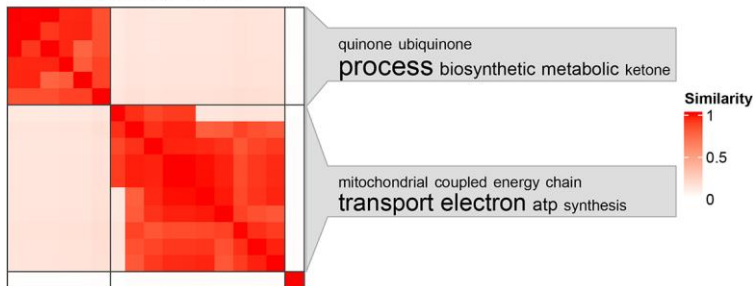


Fig. 3. Gene ontology terms and associated genes for the 10-min sampling period. The chord plot shows which genes (represented as rectangles with associated LOC ID numbers) belong to selected GO terms (full list of GO terms can be found in Supplemental Material SD.3) using colored ribbons. Genes are colored and ordered based on their log-fold change value, with a higher positive log fold change (greater expression in cold-exposed bees) having a deeper red hue, and a more negative log-fold changes having a deeper blue hue. Select differentially expressed genes of interest were highlighted based on log₂ fold-change strength, connectivity with multiple ontologies, identification of probable function (i.e. not identified as *uncharacterized loci*), and biological relevance.

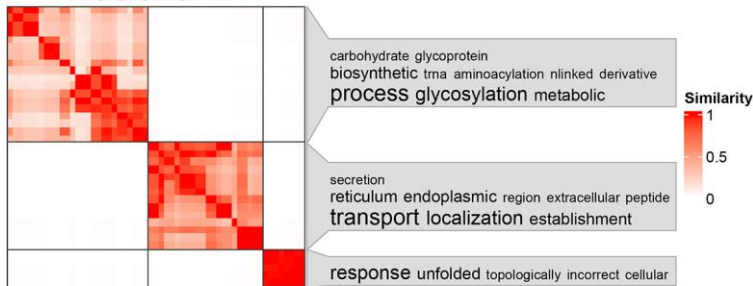
Module 1



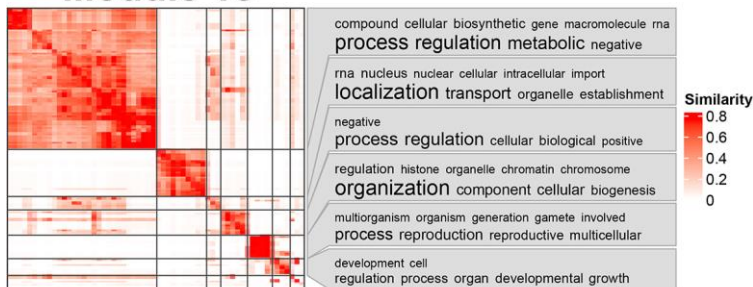
Module 8



Module 12



Module 15



Module 19

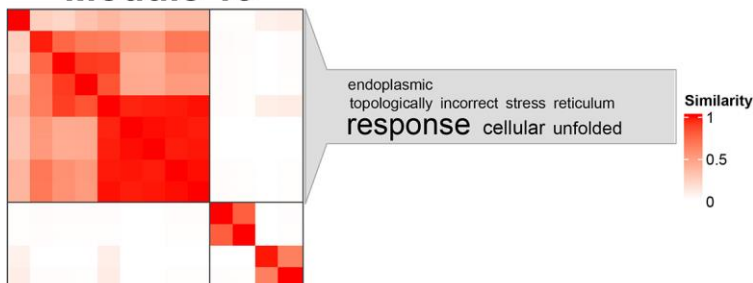


Fig. 4. Simplification and clustering of GO terms of the significant WGCNA modules best fit by the Treatment Only model. Heat maps show the similarity between clusters of similar GO terms that were identified using the binary cut method in the simplifyEnrichment package. Word clouds are used to show the most represented functional terms for each of the identified GO term clusters, with the most frequent terms having larger font sizes.

Table 1. Select genes of interest and those part of the identified GO terms from the upregulated (in cold-exposed bees) DEG list for the 10-min sampling period. The function that each gene is associated with is listed and serves as a categorical bin to help organize the genes. Within each category, the genes are ordered by log-fold change and are presented with their adjusted p-values (≤ 0.05). The gene name and LOC ID numbers are from *B. impatiens* annotations, and *D. melanogaster* homolog information is provided (FlyBase name and ID number) where applicable. Information on the WGCNA module membership is also provided. The full list of DEGs can be found in the Supplemental Material SD.1.

Function	Gene Name	BIMP 2.0 LOC ID	FlyBase Name	Flybase ID	Log2 Fold Change	Adjusted P-value	WGCNA Module Membership
Carbohydrate Metabolism	inositol monophosph atase 2	LOC100748111	CG17029	FBgn003655 1	1.61	3.91E-03	Module 4
	phosphoenol pyruvate carboxykinas e 1	LOC105680266	Pepck1	FBgn000306 7	1.24	4.50E-02	Module 4
	glycogen phosphorylas e ⁺	LOC100746613	GlyP	FBgn000450 7	1.06	9.85E-03	Module 1 ⁺
	protein phosphatase 1 regulatory subunit 3C-B	LOC100740808	Gbs-70E	FBgn003642 8	0.99	7.04E-05	Module 1 ⁺
	probable galactose-1- phosphate uridylyltransf erase	LOC100742831	Galt	FBgn026320 0	0.90	5.05E-05	Module 4
	pyruvate	LOC100743581	PyK	FBgn026738 5	0.75	1.41E-03	Module 7

	kinase						
	galactokinase	LOC100741112	N/A	N/A	0.56	3.41E-02	Module 8 ⁺
	triosephosph	LOC100741917	Tpi	FBgn0086355	0.56	2.57E-02	Module 7
	ate isomerase						
	glucose-6-phosphate isomerase	LOC100742305	Pgi	FBgn0003074	0.56	2.07E-02	Module 7
	enolase	LOC100746870	Eno	FBgn0000579	0.48	4.98E-02	Module 7
	facilitated trehalose transporter	LOC100742930	N/A	N/A	0.46	4.93E-02	Module 1 ⁺
	Tret1 ⁺						
	pyruvate carboxylase	LOC100749336	PCB	FBgn0027580	0.41	3.13E-02	Module 2
	fructose-bisphosphate aldolase	LOC100745636	Ald1	FBgn0000064	0.35	5.89E-03	Module 4
<i>Autophagy</i>	ras-related protein Rab-28	LOC100741942	N/A	N/A	0.77	3.12E-02	Module 1 ⁺
	syntaxin-17	LOC100748805	Syx17	FBgn0035540	0.59	2.22E-02	Module 4
	autophagy-related protein 9A	LOC100745297	Atg9	FBgn0034110	0.58	4.57E-03	Module 1 ⁺
	ras-related protein Rab-39B	LOC100743310	Rab39	FBgn0029959	0.58	8.54E-03	Module 4
	calcium-dependent protein kinase 16-like	LOC100741566	sqa	FBgn0259678	0.55	4.58E-02	Module 1 ⁺

	ras-related protein Rab- 21	LOC100744075	Rab21	FBgn003996 6	0.53	1.87E-02	Module 4
<i>Muscle/Neu- romuscular/Cy- toskeleton</i>	PDZ and LIM domain protein 4-like	LOC100748245	N/A	N/A	1.22	3.95E-02	Module 8 ⁺
	ankyrin-2	LOC100744281	Ank2	FBgn026178 8	0.96	1.85E-03	Module 1 ⁺
	uncharacteriz- ed	LOC100741447	Mf	FBgn003829 4	0.79	2.95E-03	Module 1 ⁺
	phosphatase and actin regulator 4	LOC100742209	CG32264	FBgn005226 4	0.67	7.64E-04	Module 1 ⁺
	uncharacteriz- ed	LOC100749880	Svil	FBgn026669 6	0.60	4.73E-02	Module 1 ⁺
	troponin C, isoallergen Bla g 6.0201	LOC100750201	TpnC41C	FBgn001334 8	0.52	1.10E-02	Module 8 ⁺
	troponin T	LOC100744757	up	FBgn000416 9	0.37	4.49E-02	Module 1 ⁺
Ion Homeostasis	sodium/potas- sium- transporting ATPase subunit beta- 1-interacting protein	LOC100744623	NKAIN	FBgn008544 2	0.70	1.75E-02	Module 1 ⁺
<i>Heat Shock Protein</i>	chaperone protein DnaJ- like	LOC105682145	N/A	N/A	0.65	2.84E-02	Module 8 ⁺
<i>Ubiquitin</i>	F-box only protein 42	LOC100741248	Fbxo42	FBgn003470 4	0.49	3.12E-02	Module 8 ⁺
	ubiquitin-	LOC100747171	CG7656	FBgn003651	0.48	2.44E-02	Module 1 ⁺

conjugating			6			
enzyme E2 R2						
E3 ubiquitin- protein ligase RNF123	LOC100741092	Kpc1	FBgn003829 6	0.41	3.41E-02	Module 1 [†]
E3 ubiquitin- protein ligase Ubr3	LOC100748893	Ubr3	FBgn026097 0	0.40	2.56E-02	Module 1 [†]
ubiquitin- conjugating enzyme E2 W	LOC100743714	CG7220	FBgn003354 4	0.38	4.15E-02	Module 4
E3 ubiquitin- protein ligase CHIP	LOC105680201	STUB1	FBgn002705 2	0.33	7.29E-03	Module 8 [†]

[†] indicates a DEG present in both the 10-min and 30-min sampling time points

^{*} indicates modules that are significantly correlated with treatment

Table 2. Select genes of interest and part of the identified GO terms from the downregulated (in cold-exposed bees) DEG list for the 10-min sampling period. The function that each gene is associated with is listed and serves as a categorical bin to help organize the genes. Within each category, the genes are ordered by log-fold change and are presented with their adjusted p-values (≤ 0.05). The gene name and LOC ID numbers are from *B. impatiens* annotations, and *D. melanogaster* homolog information is provided (FlyBase name and ID number) where applicable. Information on the WGCNA module membership is also provided. The full list of DEGs can be found in the Supplemental Material SD.1.

Function	Gene Name	BIMP 2.0 LOC ID	FlyBase Name	Flybase ID	Log2 Fold Change	Adjusted P-value	WGCNA Module Membership
Purine Metabolism	phosphoribosylformyl glycinamide synthase	LOC100749333	Pfas	FBgn00000 52	-2.03	1.47E-03	Module 14
	amidophosphoribosyl transferase	LOC100749989	Prat2	FBgn00411 94	-1.91	1.38E-02	Module 14
	GART trifunctional enzyme	LOC100745134	Gart	FBgn00000 53	-1.61	5.89E-03	Module 14
	serine hydroxymethyltransfe rase, cytosolic	LOC100741844	Shmt	FBgn00298 23	-1.10	1.59E-02	Module 14
	xanthine dehydrogenase	LOC100745073	ry	FBgn00033 08	-1.06	3.04E-02	Module 14
	adenine phosphoribosyltransfe rase	LOC100746243	Aprt	FBgn00001 09	-0.93	1.75E-02	Module 12 ⁺
	phospholipase A1 ⁺	LOC100747885	CG6295	FBgn00394 71	-4.28	6.61E-03	Module 12 ⁺
Lipid Metabolism	pancreatic triacylglycerol lipase- like	LOC100741829	CG5966	FBgn00298 31	-1.24	2.94E-02	Module 14

	probable enoyl-CoA hydratase, mitochondrial	LOC100746130	Echs1	FBgn0033879	-0.95	2.63E-02	Module 14
	putative fatty acyl-CoA reductase CG8306	LOC100742685	CG8306	FBgn0034142	-0.90	4.38E-02	Module 11
	short-chain specific acyl-CoA dehydrogenase, mitochondrial	LOC100741796	Arc42	FBgn0038742	-0.62	3.70E-02	Module 14
	very-long-chain 3-oxoacyl-CoA reductase	LOC100744619	spidey	FBgn0029975	-2.11	1.12E-02	Module 14
<i>Heat Shock Protein</i>	protein lethal(2)essential for life	LOC100742443	N/A	N/A	-1.89	5.77E-04	Module 10
	heat shock protein 60A	LOC100748180	Hsp60A	FBgn0015245	-0.94	8.51E-03	Module 15 ⁺
	97 kDa heat shock protein	LOC100742698	Hsc70Cb	FBgn0026418	-0.72	2.34E-02	Module 5
	dnaJ homolog subfamily C member 1	LOC100748155	CG7556	FBgn0030990	-0.34	3.12E-02	Module 12 ⁺
<i>Endoplasmic Reticulum Stress/Protein Folding</i>	mesencephalic astrocyte-derived neurotrophic factor homolog	LOC100740746	Manf	FBgn0027095	-0.92	2.64E-02	Module 19 ⁺
	protein disulfide-isomerase A3	LOC100749925	ERp60	FBgn0033663	-0.89	6.61E-03	Module 19 ⁺
	endoplasmic reticulum resident protein 29	LOC100741311	wbl	FBgn0004003	-0.87	4.74E-02	Module 19 ⁺

	derlin-2	LOC100740663	Der-2	FBgn0038438	-0.52	1.7E-02	Module 19 [†]
<i>Cytoskeleton</i>	Septin-2	LOC100750127	Septin2	FBgn0014029	-0.49	1.56E-02	Module 5

[†] indicates a DEG present in both the 10-min and 30-min sampling time points

^{*} indicates modules that are significantly correlated with treatment

Table 3. Identifying the best fit model for changes in gene expression modules between treatments and across recovery time using AIC values. Each model is displayed along with their degrees of freedom and AIC value for each of the significant modules. Information showing the significance of each module tested (estimated using the Satterthwaite approximation) against all the models can be found in Supplementary Material SI (Table.SI.3A-3E).

Models	df	Modules 1 AIC	Module 8 AIC	Module 12 AIC	Module 15 AIC	Module 19 AIC
Full	12	-60.45820	-59.03047	-55.17738	-53.79321	-54.16898
Treatment Only	4	-93.38414 [‡]	-99.02165 [‡]	-97.20972 [‡]	-95.02896 [‡]	-99.48840 [‡]
Time Only	7	-72.97647	-78.56377	-70.78709	-74.66283	-69.04966
Treatment and Time	8	-71.31733	-75.25834	-71.82427	-74.07057	-72.51036
Numb						
ers of		1283	361	507	571	57
Genes						

[‡] indicates the best fit module and lowest AIC value

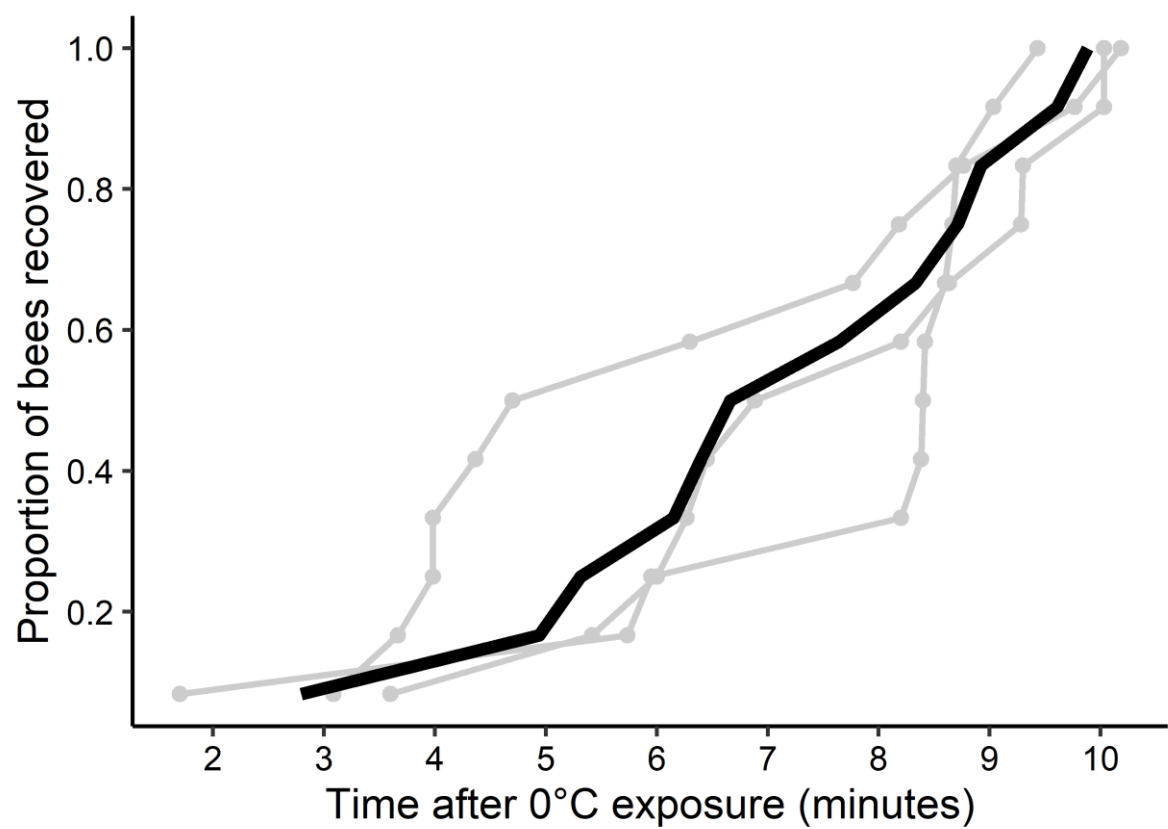


Fig. S1. Worker bees recovered from chill coma within 10 minutes of returning to room temperature conditions from the cold exposure. The black line represents the proportion of bees recovered from chill coma over time across all bees, which came from three different colonies (grey lines)

Table S1. Complete list of samples retained and used for all the analyses. Data includes the colony of origin for sample (Colony ID), if this individual was used as a control or subjected to the cold shock (Treatment), the time following the acute cold shock when each individual was sampled (Time), and how many sequenced reads were generated for each individual (Read Count).

Sample ID	Colony ID	Treatment	Time	Read Count
C00ECK5	Daisy	Control	00m	12515608
C00ECK6	Daisy	Control	00m	15768802
C00ECK40	Dahlia	Control	00m	15321093
C00ECK77	Violet	Control	00m	10499414
C10ECK7	Daisy	Control	10m	12172138
C10ECK8	Daisy	Control	10m	15338675
C10ECK43	Dahlia	Control	10m	16262294
C10ECK44	Dahlia	Control	10m	15811361
C10ECK45	Dahlia	Control	10m	15274922
C10ECK79	Violet	Control	10m	8381891
C10ECK80	Violet	Control	10m	9852370
C10ECK81	Violet	Control	10m	10033816
C30ECK10	Daisy	Control	30m	12608893
C30ECK11	Daisy	Control	30m	13013365
C30ECK12	Daisy	Control	30m	20420624
C30ECK46	Dahlia	Control	30m	13672231
C30ECK47	Dahlia	Control	30m	12248873

C30ECK48	Dahlia	Control	30m	15442226
C30ECK83	Violet	Control	30m	12651098
C30ECK84	Violet	Control	30m	9641143
C120ECK13	Daisy	Control	120m	12293423
C120ECK14	Daisy	Control	120m	14074181
C120ECK15	Daisy	Control	120m	13714552
C120ECK49	Dahlia	Control	120m	10104000
C120ECK50	Dahlia	Control	120m	13467508
C120ECK51	Dahlia	Control	120m	16511195
C120ECK85	Violet	Control	120m	13052129
C120ECK87	Violet	Control	120m	14761455
C720ECK16	Daisy	Control	720m	9092337
C720ECK17	Daisy	Control	720m	15289841
C720ECK52	Dahlia	Control	720m	12468667
C720ECK53	Dahlia	Control	720m	14248492
C720ECK54	Dahlia	Control	720m	14988727
C720ECK88	Violet	Control	720m	12279638
C720ECK89	Violet	Control	720m	11271166
T00ECK22	Daisy	Cold	00m	10068522
T00ECK23	Daisy	Cold	00m	11938929
T00ECK24	Daisy	Cold	00m	12296678
T00ECK59	Dahlia	Cold	00m	10544331
T00ECK60	Dahlia	Cold	00m	11556593

T00ECK94	Violet	Cold	00m	11519041
T00ECK95	Violet	Cold	00m	16149799
T00ECK96	Violet	Cold	00m	13513752
T10ECK25	Daisy	Cold	10m	13900347
T10ECK26	Daisy	Cold	10m	13282204
T10ECK27	Daisy	Cold	10m	12680738
T10ECK61	Dahlia	Cold	10m	12102052
T10ECK62	Dahlia	Cold	10m	13209112
T10ECK97	Violet	Cold	10m	15784056
T10ECK99	Violet	Cold	10m	12462735
T30ECK28	Daisy	Cold	30m	9557892
T30ECK29	Daisy	Cold	30m	14474658
T30ECK64	Dahlia	Cold	30m	13705275
T30ECK65	Dahlia	Cold	30m	10739819
T30ECK66	Dahlia	Cold	30m	8514524
T30ECK100	Violet	Cold	30m	12756536
T30ECK101	Violet	Cold	30m	13192165
T30ECK102	Violet	Cold	30m	13238234
T120ECK31	Daisy	Cold	120m	13496414
T120ECK32	Daisy	Cold	120m	13814344
T120ECK33	Daisy	Cold	120m	8424910
T120ECK67	Dahlia	Cold	120m	10566911
T120ECK68	Dahlia	Cold	120m	13600540
T120ECK69	Dahlia	Cold	120m	11539789
T120ECK103	Violet	Cold	120m	13889475
T120ECK104	Violet	Cold	120m	12179560
T120ECK105	Violet	Cold	120m	11451612
T720ECK34	Daisy	Cold	720m	10724752
T720ECK36	Daisy	Cold	720m	12715841
T720ECK70	Dahlia	Cold	720m	17760658
T720ECK72	Dahlia	Cold	720m	13078227
T720ECK106	Violet	Cold	720m	11522471
T720ECK107	Violet	Cold	720m	15862864
T720ECK108	Violet	Cold	720m	8514585

Table S2. List of modules identified through the WGCNA analysis. The number of genes assigned to each module are presented (Number of Genes) along with the number of DEGs (Number of DEGS) assigned to each module for the 10-min and 30-min time series (The only time series will differential expression). The hub gene for each module is also provided (Hub Gene) with the LOC ID (BIMP 2.0 LOC ID) and the gene name (Gene Name) presented.

WGCNA Module ID	Number of Genes	Number of DEGs		BIMP 2.0 LOC ID	Hub Gene Gene Name
		10-min	30-min		
Module 1	1283	104	28	LOC100747735	nuclear receptor coactivator 2
Module 2	1260	5	NA	LOC100747609	histone acetyltransferase p300
Module 3	871	2	NA	LOC100745841	39S ribosomal protein L13, mitochondrial
Module 4	1241	59	2	LOC100742828	40S ribosomal protein S16
Module 5	427	7	2	LOC100743946	malignant T-cell-amplified sequence 1 homolog
Module 6	70	NA	NA	LOC100742851	chymotrypsin-1
Module 7	489	19	NA	LOC100747829	cytochrome b-c1 complex subunit 2, mitochondrial
Module 8	361	41	NA	LOC100747551	complex III assembly factor LYRM7
Module 9	192	5	NA	LOC105680328	uncharacterized
Module 10	165	1	4	LOC100749321	uncharacterized
Module 11	160	2	NA	LOC100742591	uncharacterized
Module 12	508	27	5	LOC100748209	mannose-1-phosphate guanyltransferase beta
Module13	272	5	1	LOC100741439	vascular endothelial growth factor receptor 1
Module 14	557	60	1	LOC100745424	uncharacterized
Module 15	571	14	1	LOC100747231	ras GTPase-activating protein-binding protein 2
Module 16	308	1	NA	LOC105682033	uncharacterized
Module 17	45	NA	NA	LOC100740515	probable chitinase 2
Module 18	1487	8	7	LOC100740171	protein couch potato
Module 19	57	4	NA	LOC100740746	mesencephalic astrocyte-derived neurotrophic factor homolog
Module 20	764	5	NA	LOC100743548	cytochrome b-c1 complex subunit 6, mitochondrial

Table S3A. Results from the linear mixed model (R package lme4) used to test for associations between Module 1 and experimental factors. Bolded p-values indicate significance (< 0.05).

	Module1				Module1				Module1						
Predictors	Estimates	std.	Error	Statistic	p	Estimates	std.	Error	Statistic	p	Estimates	std.	Error	Statistic	p
(Intercept)	-0.065	0.054	-1.203	0.234		-0.032	0.021	-1.517	0.177		-0.055	0.038	-1.447	0.155	
Treatment [Treatment]	0.080	0.065	1.233	0.222		0.060	0.026	2.272	0.026		0.064	0.026	2.440	0.017	
Time [10m]	0.048	0.065	0.730	0.468							0.074	0.044	1.695	0.095	
Time [120m]	0.060	0.065	0.925	0.359							-0.008	0.043	-0.181	0.857	
Time [30m]	-0.009	0.065	-0.146	0.885							0.028	0.043	0.659	0.512	
Time [720m]	0.054	0.067	0.809	0.422							0.006	0.044	0.141	0.888	
Treatment [Treatment] × Time [10m]	0.063	0.086	0.741	0.461											
Treatment [Treatment] × Time [120m]	-0.125	0.083	-1.500	0.139											
Treatment [Treatment] × Time [30m]	0.080	0.084	0.956	0.343											
Treatment [Treatment] × Time [720m]	-0.091	0.087	-1.053	0.296											
Random Effects															
σ²	0.01					0.01					0.01				
τ₀₀	0.00	Colony				0.00	Colony				0.00	Colony			
ICC	0.02					0.02					0.01				
N	3	Colony				3	Colony				3	Colony			
Observations	74					74					74				
Marginal R² / Conditional R²	0.238 / 0.253					0.065 / 0.080					0.124 / 0.136				

Table S3B. Results from the linear mixed model (R package lme4) used to test for associations between Module 8 and experimental factors. Bolded p-values indicate significance (< 0.05).

Predictors	Module8				Module8				Module8			
	Estimates	std. Error	Statistic	p	Estimates	std. Error	Statistic	p	Estimates	std. Error	Statistic	p
(Intercept)	-0.064	0.060	-1.069	0.297	-0.028	0.034	-0.814	0.479	-0.039	0.046	-0.852	0.418
Treatment [Treatment]	0.091	0.065	1.403	0.166	0.053	0.025	2.111	0.038	0.052	0.025	2.074	0.042
Time [10m]	-0.037	0.065	-0.576	0.567					-0.017	0.042	-0.406	0.686
Time [120m]	0.076	0.065	1.173	0.245					0.040	0.040	0.996	0.323
Time [30m]	0.077	0.065	1.200	0.235					0.036	0.041	0.879	0.382
Time [720m]	0.053	0.066	0.801	0.426					-0.011	0.042	-0.254	0.800
Treatment [Treatment] × Time [10m]	0.060	0.085	0.705	0.483								
Treatment [Treatment] × Time [120m]	-0.057	0.082	-0.691	0.492								
Treatment [Treatment] × Time [30m]	-0.070	0.083	-0.842	0.403								
Treatment [Treatment] × Time [720m]	-0.115	0.086	-1.348	0.183								
Random Effects												
σ ²	0.01				0.01				0.01			
τ ₀₀	0.00 _{Colony}				0.00 _{Colony}				0.00 _{Colony}			
ICC	0.19				0.18				0.18			
N	3 _{Colony}				3 _{Colony}				3 _{Colony}			
Observations	74				74				74			
Marginal R ² / Conditional R ²	0.140 / 0.299				0.048 / 0.217				0.086 / 0.252			

Table S3C. Results from the linear mixed model (R package lme4) used to test for associations between Module 12 and experimental factors. Bolded p-values indicate significance (< 0.05).

Predictors	Module12				Module12				Module12			
	Estimates	std. Error	Statistic	p	Estimates	std. Error	Statistic	p	Estimates	std. Error	Statistic	p
(Intercept)	0.091	0.057	1.597	0.116	0.039	0.023	1.735	0.147	0.058	0.039	1.494	0.145
Treatment [Treatment]	-0.128	0.068	-1.892	0.063	-0.075	0.026	-2.941	0.004	-0.078	0.026	-2.984	0.004
Time [10m]	-0.036	0.068	-0.525	0.602					-0.029	0.044	-0.670	0.505
Time [120m]	-0.056	0.068	-0.827	0.411					0.009	0.042	0.223	0.824
Time [30m]	-0.048	0.068	-0.704	0.484					-0.036	0.043	-0.847	0.400
Time [720m]	-0.097	0.069	-1.401	0.166					-0.026	0.044	-0.590	0.557
Treatment [Treatment] × Time [10m]	-0.007	0.089	-0.084	0.934								
Treatment [Treatment] × Time [120m]	0.111	0.086	1.280	0.205								
Treatment [Treatment] × Time [30m]	0.006	0.087	0.071	0.943								
Treatment [Treatment] × Time [720m]	0.126	0.090	1.404	0.165								
Random Effects												
σ ²	0.01				0.01				0.01			
τ ₀₀	0.00 _{Colony}				0.00 _{Colony}				0.00 _{Colony}			
ICC	0.04				0.04				0.03			
N	3 _{Colony}				3 _{Colony}				3 _{Colony}			
Observations	74				74				74			
Marginal R ² / Conditional R ²	0.171 / 0.204				0.102 / 0.139				0.121 / 0.150			

Table S3D. Results from the linear mixed model (R package lme4) used to test for associations between Module 15 and experimental factors. Bolded p-values indicate significance (< 0.05).

Predictors	Module15				Module15				Module15			
	Estimates	std. Error	Statistic	p	Estimates	std. Error	Statistic	p	Estimates	std. Error	Statistic	p
(Intercept)	0.091	0.056	1.604	0.114	0.036	0.019	1.917	0.077	0.055	0.036	1.502	0.140
Treatment [Treatment]	-0.123	0.069	-1.783	0.079	-0.069	0.026	-2.641	0.010	-0.069	0.026	-2.671	0.009
Time [10m]	-0.028	0.069	-0.405	0.687					-0.002	0.043	-0.050	0.960
Time [120m]	-0.123	0.069	-1.777	0.080					-0.064	0.042	-1.541	0.128
Time [30m]	-0.077	0.069	-1.115	0.269					-0.038	0.042	-0.891	0.376
Time [720m]	-0.011	0.071	-0.149	0.882					0.027	0.044	0.621	0.537
Treatment [Treatment] × Time [10m]	0.032	0.090	0.359	0.721								
Treatment [Treatment] × Time [120m]	0.096	0.088	1.093	0.278								
Treatment [Treatment] × Time [30m]	0.061	0.089	0.683	0.497								
Treatment [Treatment] × Time [720m]	0.058	0.092	0.630	0.531								
Random Effects												
σ ²	0.01				0.01				0.01			
τ ₀₀	0.00	Colony			0.00	Colony			0.00	Colony		
ICC	0.00				0.00				0.00			
N	3	Colony			3	Colony			3	Colony		
Observations	74				74				74			
Marginal R ² / Conditional R ²	0.165	0.167			0.087	0.087			0.157	0.160		

Table S3E. Results from the linear mixed model (R package lme4) used to test for associations between Module 19 and experimental factors. Bolded p-values indicate significance (< 0.05).

Predictors	Module19				Module19				Module19			
	Estimates	std. Error	Statistic	p	Estimates	std. Error	Statistic	p	Estimates	std. Error	Statistic	p
(Intercept)	0.112	0.056	1.994	0.050	0.046	0.018	2.517	0.014	0.058	0.037	1.576	0.120
Treatment [Treatment]	-0.171	0.069	-2.483	0.016	-0.088	0.025	-3.467	0.001	-0.090	0.026	-3.413	0.001
Time [10m]	-0.079	0.069	-1.140	0.258					-0.020	0.044	-0.469	0.640
Time [120m]	-0.095	0.069	-1.375	0.174					-0.003	0.042	-0.065	0.949
Time [30m]	-0.063	0.069	-0.917	0.362					-0.019	0.043	-0.445	0.658
Time [720m]	-0.059	0.071	-0.832	0.409					-0.009	0.044	-0.208	0.836
Treatment [Treatment] × Time [10m]	0.090	0.090	0.993	0.325								
Treatment [Treatment] × Time [120m]	0.153	0.088	1.735	0.087								
Treatment [Treatment] × Time [30m]	0.061	0.089	0.686	0.495								
Treatment [Treatment] × Time [720m]	0.072	0.091	0.785	0.435								
Random Effects												
σ ²	0.01				0.01				0.01			
τ ₀₀	0.00 _{Colony}				0.00 _{Colony}				0.00 _{Colony}			
N	3 _{Colony}				3 _{Colony}				3 _{Colony}			
Observations	74				74				74			
Marginal R ² / Conditional R ²	0.170 / NA				0.141 / NA				0.139 / NA			

Table S4. Dataset

Available for download at
<https://journals.biologists.com/jeb/article-lookup/doi/10.1242/jeb.247040#supplementary-data>